Ministry Approved
2013

Compulsory Education Textbook

Mathematics

SHUXUE

Grade 3 Volume 1

北京师范大学出版社

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8.3

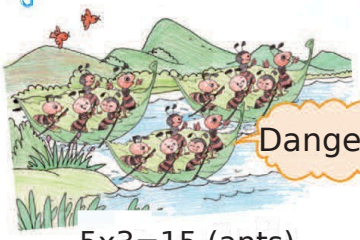
Dear children,

Wise Old Man

Now in Grade 3, what new understanding of mathematics do you have?

Mathematics is made up of countless mathematical stories.

Taoqi

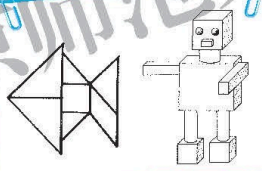


5x3=15 (ants)

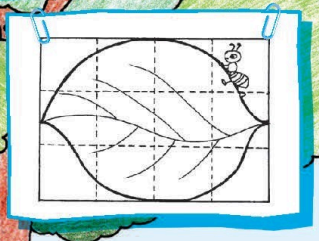
4x4=16 (ants)
16-1=15 (ants)

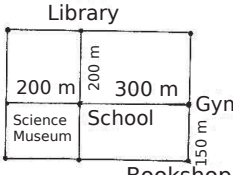
Mathematics is like a big tree: the more you learn, the bigger and leafier it grows.

Xiaoxiao



45+45=90
100-19=81
7×6=42
25÷4=6……1





Woof! Maths is wonderful!

Clever Dog

Are you ready, children? Let's step into the Kingdom of Mathematics together. It will bring you even more surprises and joy!

Your friends, the editors

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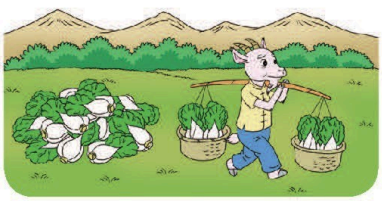

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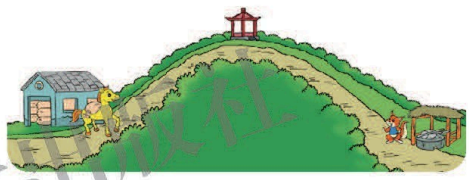

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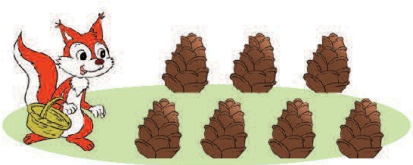

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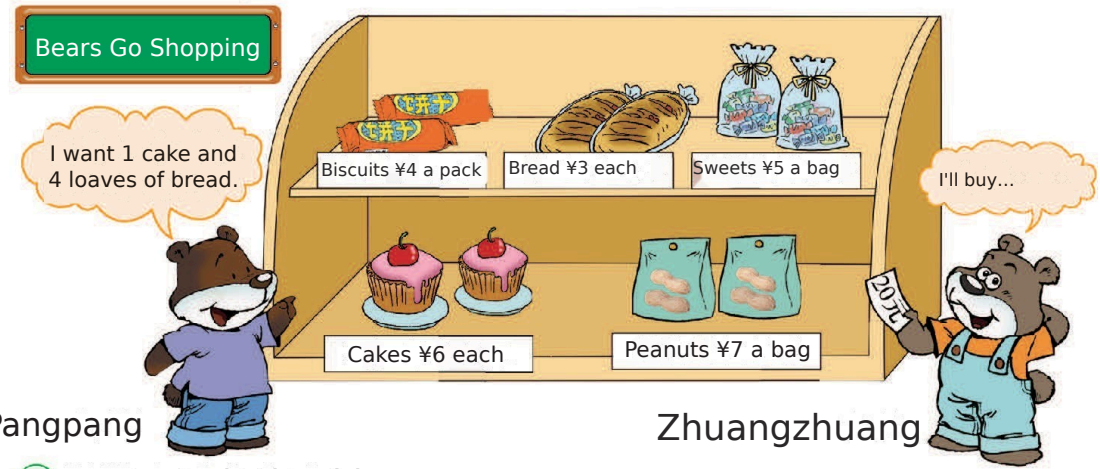
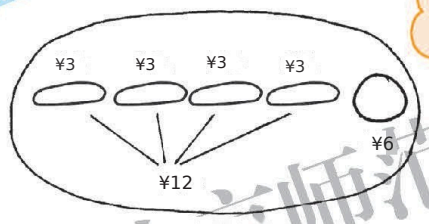
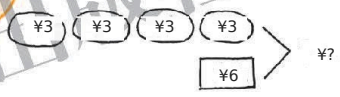
Unit 1 Mixed Operations



How much should Pangpang pay?

First work out the cost of 4 loaves, then...





3x4=12 (yuan)
12+6=18 (yuan)

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Answer: _____ .

Do you understand Taoqi's and Xiaoxiao's number sentences?

I take the money for the bread and add...

3x4+6

6+3×4

I take the money for the cake and add...

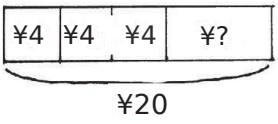
With addition and multiplication, what do you work out first, and what next?

3x4+6
=12+6
=

6+3×4
= 6+12
=

Try It

Zhuangzhuang has ¥20. How much change should he get after buying 3 packs of biscuits?



3×4=12 (yuan)
20-12=8 (yuan)

With subtraction and multiplication, what comes first? What next?

20-3×4
=
=

Answer: _____ .

Use the shopping picture to explain each number sentence, then calculate.

$20-2\times7$
=
=

$5\times3+4$
=
=

$4\times6-20$
=
=

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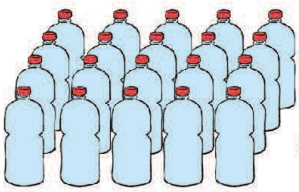
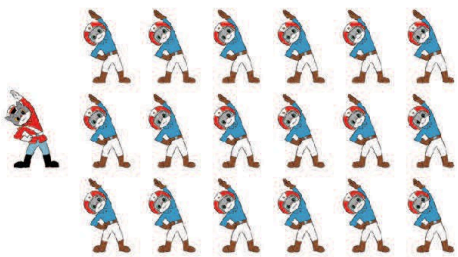
Practice

1. Discuss, then write a number sentence and calculate.

How many bottles are there in all?



How many police cats altogether?

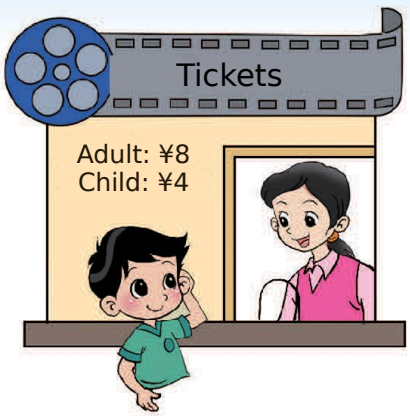


2. Say what to work out first and what next, then calculate.

$4\times6+25$　　$54+36-18$　　$42+8\times4$

$9\times8+22$　　$100-75+25$　　$33+7\times3$

3.



(1) Taoqi has ¥50. How much change should he get after buying 8 child tickets?

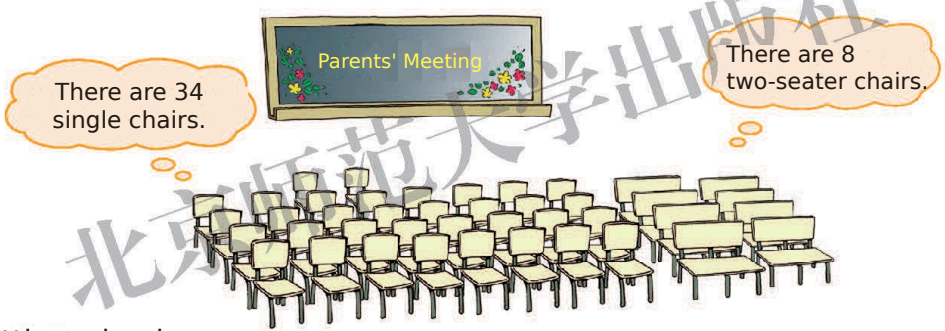
(2) Say what the number sentences below mean in this situation.

$8\times2+4$	$50-4\times5$	$8\times7-50$

4.

$50-5\times8$	$7\times6-30$	$93-3\times9$
$8\times9-34$	$20-2\times6$	$46+4\times7$

5. How many people can sit in total?



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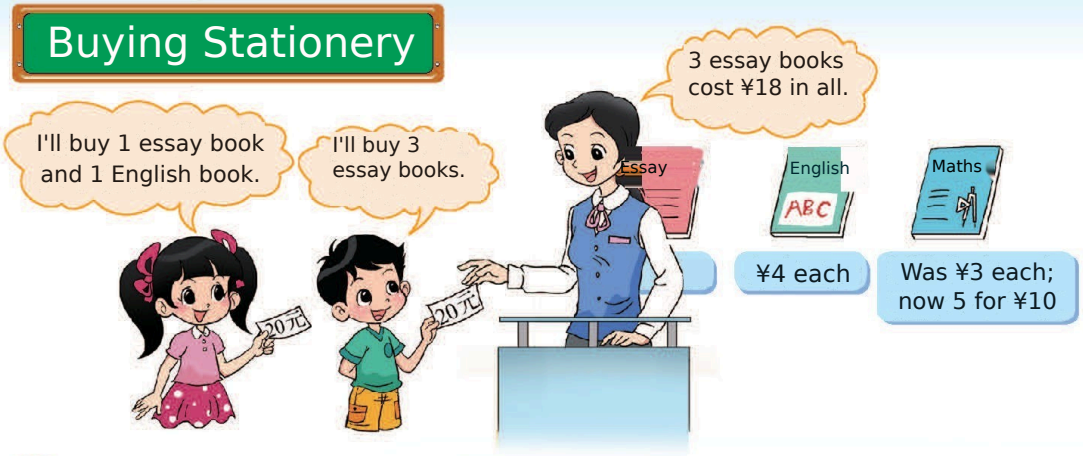
6. Putting wheels on cars.

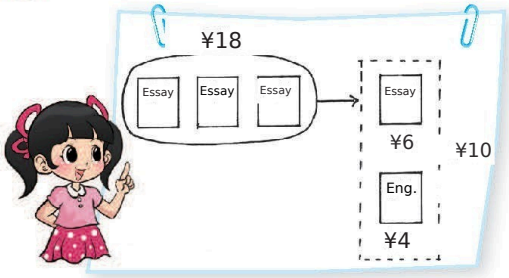
(1) If wheels are put on 5 cars, how many wheels are left?

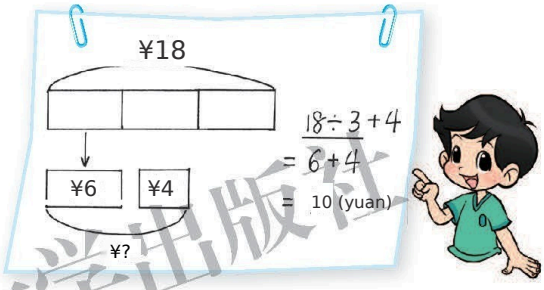
(2) If wheels are put on 8 cars, how many more wheels are needed?

Buying Stationery



How much money does Xiaoxiao need in all?





Answer: _____。

How much cheaper is each maths book now? Talk about it, then fill in.

10÷5=2 (yuan)

□○□=□ (yuan)

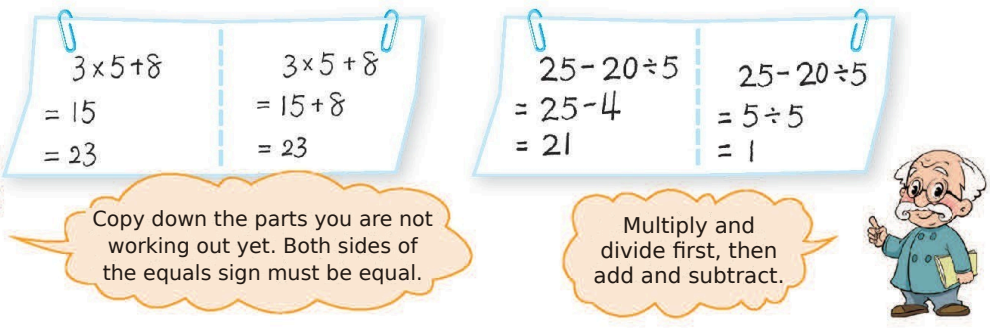
3−10÷5

=□−□

=□ (yuan)

Answer: _____。

Which method is right? Talk about it.



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Try It

Last time I bought 2 pens and spent ¥16. I paid too much.

Pens are ¥7 each

Ballpoint pens are ¥6 each

Mechanical pencils: original price ¥5 each, current price 4 for ¥12

How much more did each of Taoqi's pens cost than a pen in the stationery shop now?

First work out how much each pen cost Taoqi last time...

$16 \div 2 =$

$16 \div 2 - 7 =$

Answer: _____ .

Use the situation above to explain the number sentences below, then calculate.

$3 \times 7 + 12$ $5 - 12 \div 4$

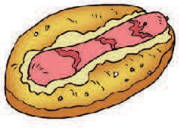
Practice

1.

Buy 2 cream buns for ¥14

Buy 1 chocolate bun for ¥5

Buy 3 ham buns for ¥12



(1) How much does it cost in total to buy 1 cream bun and 1 chocolate bun?

(2) How much more does 1 chocolate bun cost than 1 ham bun?

2. Say what to work out first and what next, then calculate.

$4 + 24 \div 8$	$72 \div 8 - 3$	$5 - 8 \div 4$
$45 - 5 \times 4$	$86 - 6 \times 8$	$9 + 81 \div 9$

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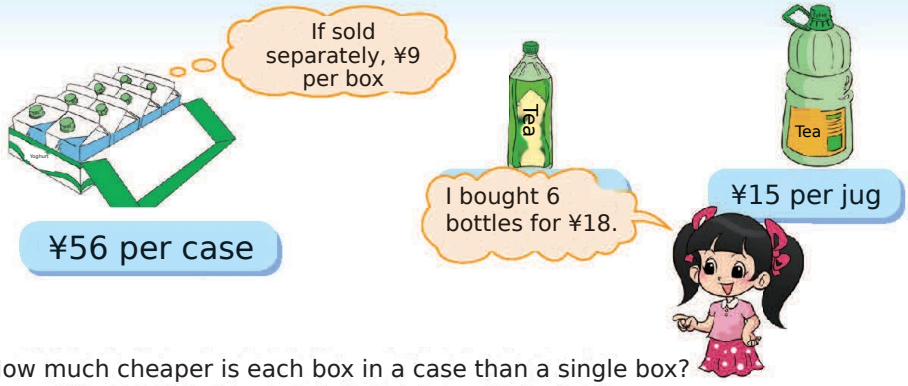
3.

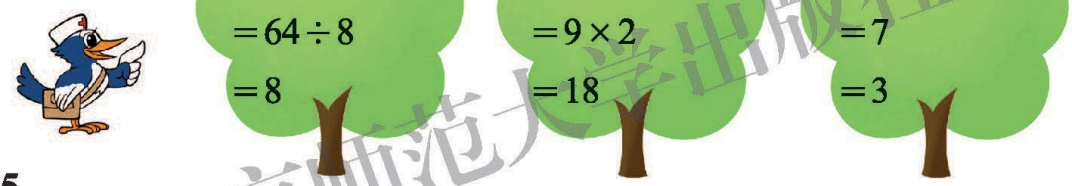


(1) How much cheaper is each box in a case than a single box?

(2) Use the situation above to explain the number sentences below, then calculate.

$9+18\div6$	$3\times9+15$

4. Forest Doctor



5.

$45\div5+14$	$9\times7-30$	$45+15\div3$
$72\div9-8$	$20+6\times6$	$15-48\div8$

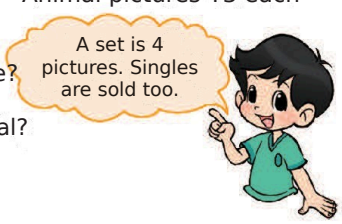
6.

Landscape pictures ¥20 per set Flower pictures ¥2 each Animal pictures ¥3 each

(1) How much more is a landscape picture than a flower picture?

(2) How much do a landscape set and an animal set cost in total?

(3) Ask another maths question and try to solve it.



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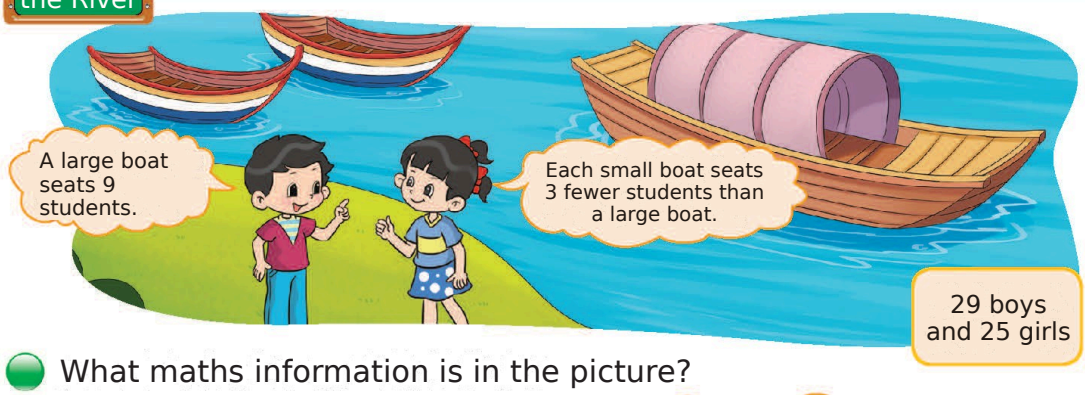
Crossing the River



What maths information is in the picture?

I know there are 29 boys and 25 girls.

I know a large boat seats 9 students, and a small boat can seat...

If everyone takes large boats, how many boats are needed? Can you follow their discussion?

Is it all right to do it this way?

$29 + 25 \div 9$
$= 54 \div 9$
$= 6$ (boats)

No. Written this way, you have to do the division first...

Ask "( )" to help: work out what is inside the brackets first.

$(29 + 25) \div 9$
$= \square \div \square$
$= \square$ (boats)

Brackets can change the order of operations. They were first used by the Dutchman Girard.

Answer: _____ .

If all 54 students take small boats, how many boats are needed? Can you work it out?

First find how many people a small boat seats, then...

I'll try writing one number sentence.

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Try It

70 people in all. Fill the big bus first; the rest take minibuses. What is the least number of minibuses needed?

"Seats 46 passengers" means...

$(70-46)\div 8$

$=$

First find how many people are left, then...

Answer: _____ .

Look and say: what other problems can be solved with $(70-46)\div 8$?

I have read 46 pages and want to finish the rest in 8 days. On average...

70 pages

¥46 each

¥8 each

I had ¥70 and bought 1 schoolbag. With the money left...

Practice

1. In Class 3(1), 18 boys and the same number of girls take part in the formation and group gymnastics performances.

In the formation performance, we stand in 4 rows.

In the group gymnastics performance, 3 boys and 3 girls form a pattern.

(1) What maths information can you find?

(2) In the formation, on average, how many people stand in each row?

(3) In the group gymnastics, how many patterns can 36 people form?

2. Say what to work out first and what next, then calculate.

$5\times(36-29)$ $(46+17)\div 7$ $81\div(16-7)$

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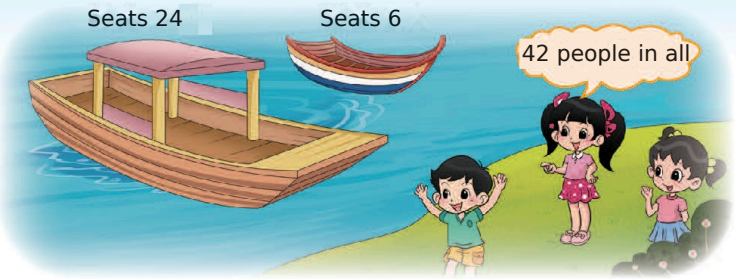
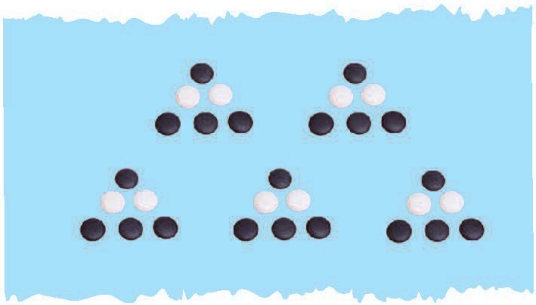
3. Fill the big boat first and put the rest in small boats. What is the least number of small boats needed?



4. Look and say: what problems can be solved with (4+2)×5?



5.

Juice

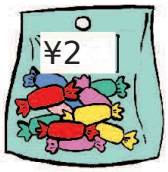
Cup noodles ¥4

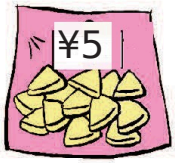
Noodles ¥6 per bag

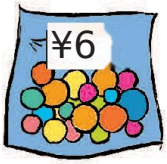
Rice ¥64 per bag

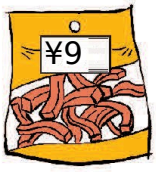
(1) Uncle Li has ¥100. He buys 1 bag of rice. How many bags of noodles can he buy with the rest?
(2) Taoqi spent ¥19 on 4 cups of noodles and 1 bottle of juice. How much is a bottle of juice?
(3) Ask another maths question and try to solve it.

6. Xiaoxiao bought one kind of sweets. She paid ¥20 and got ¥2 change. Which kind might she have bought? How many bags did she buy?









7. Add brackets to make the number sentences below true.

$15-6\div3=3$ $2\times5+3=16$ $15\div6-3=5$

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Exercise 1

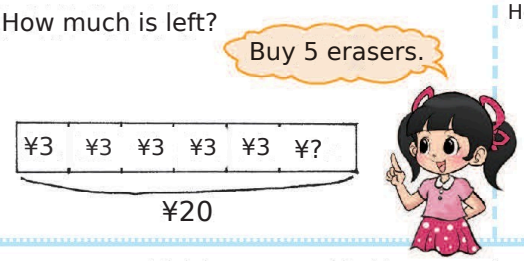
1. Discuss and calculate.

How much do they cost altogether?

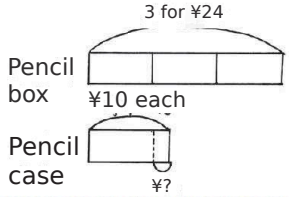
6 in each row. How many rows can they make in all?

12 rabbits 18 rabbits

How much is left?



How much more is 1 pencil case than 1 pencil box?



2. Say what to work out first and what next, then calculate.

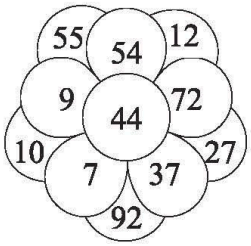
$49-3\times8$	$14+6\times4$	$3\times8-16$	$(35+28)\div7$
$(99-78)\div3$	$100-(72+28)$	$57-34-14$	$28+65-85$

3. Colour the answers below red in the picture on the right.

$(6+57)\div9$ $\quad$ $63-27\div3$

$8\times(69-60)$ $\quad$ $99-(23+32)$

$(100-46)\div6$ $\quad$ $25+3\times4$



4. How many points did the Star Football Team get in all?

5. Who bought the cheaper drink, and how much cheaper is each bottle?

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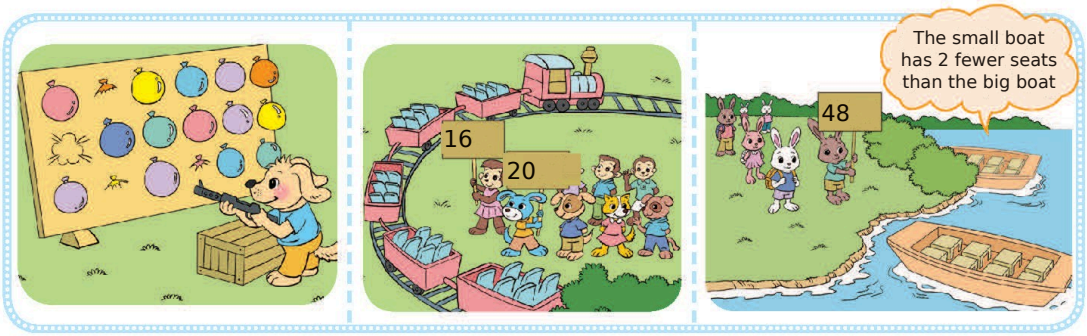
6. How many times as many black swans are there as white swans?

45 black swans.

There are 36 fewer white swans than black swans.

7. Animal Fun Park



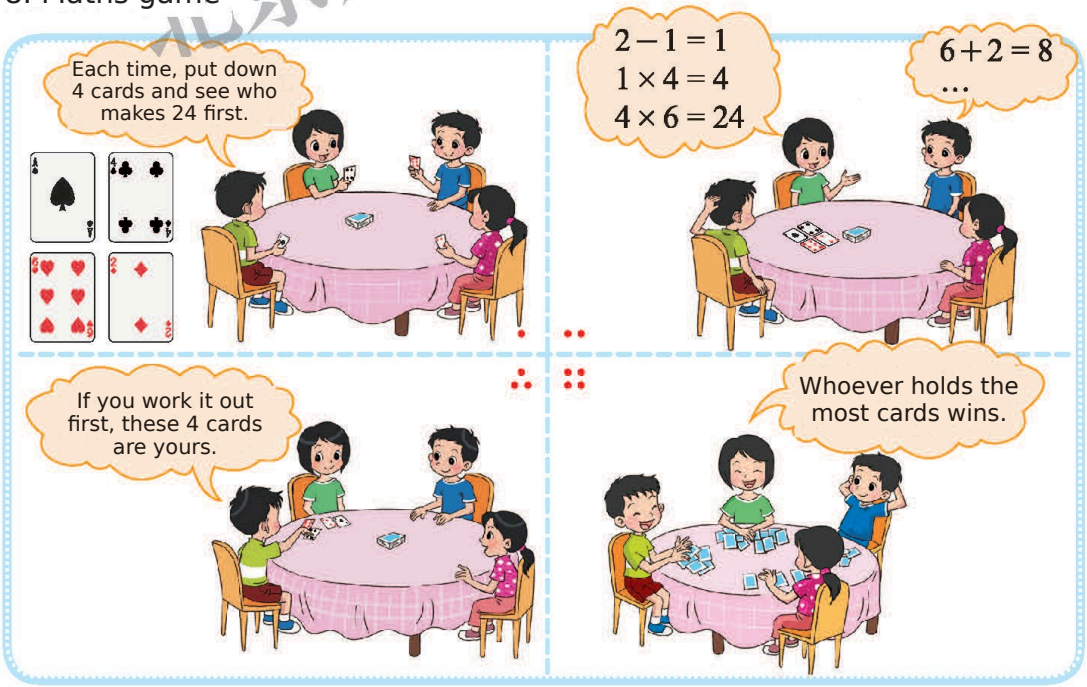
(1) What maths information do you get from the picture?

(2) How many carriages can the dogs and monkeys fill?

(3) Use the picture to explain the number sentences, then calculate.

$3 \times 6 - 4$ $48 \div (8 - 2)$

8. Maths game



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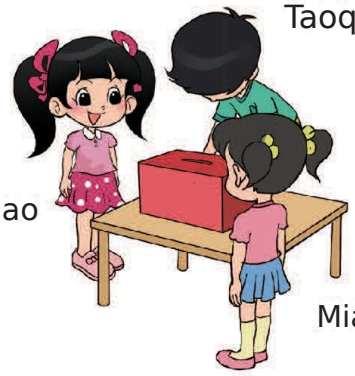
Unit 2 Observing Objects

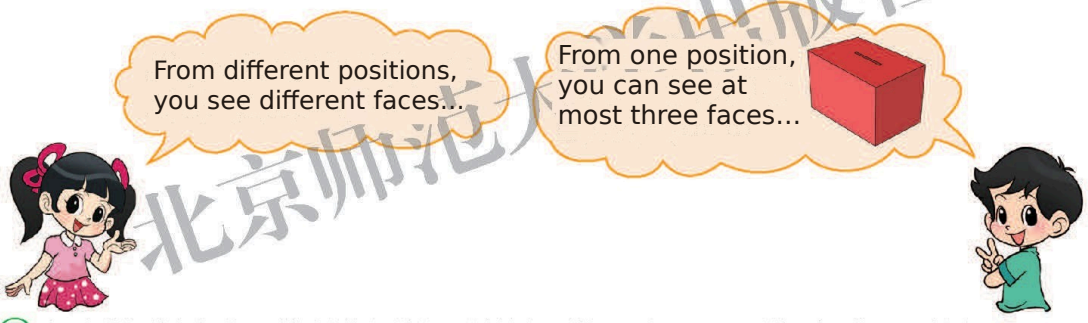
Look (1)



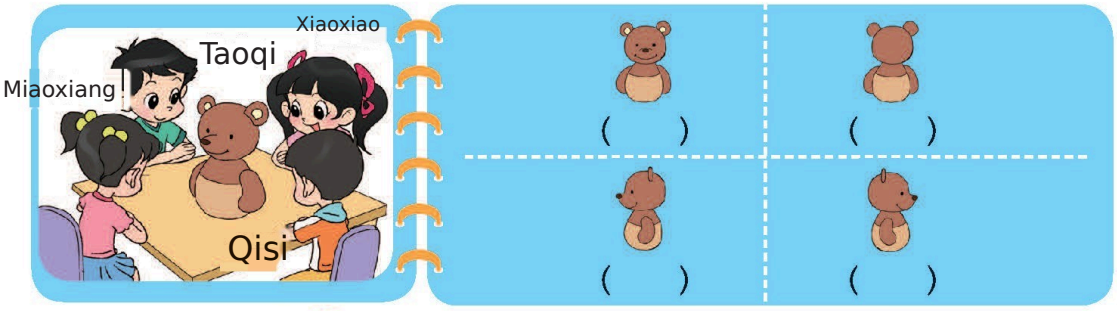
Who sees each of the three pictures below? Think and match.

Xiaoxiao Miaoxiang Taoqi

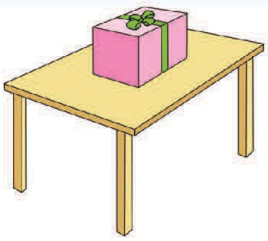
How many faces can you see at most from each position? Think first, then look.



Which classmate sees each of the four pictures below? Think first, then look.

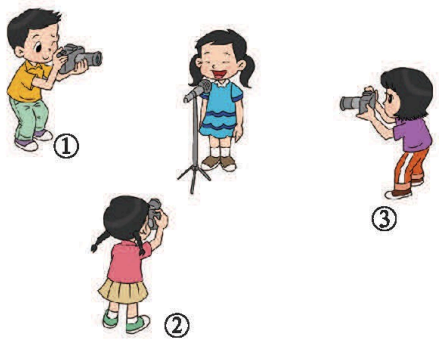


Practice

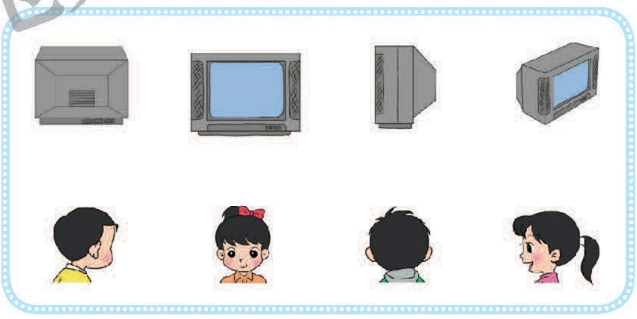


1. Put a gift box on a table and look at it from different places. How many faces can you see at most? Talk about it.

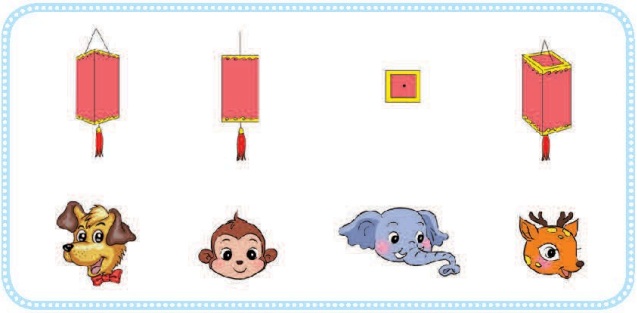
2. Think: which young reporter took each of the three photos below? Write their numbers in the brackets under the photos.



3. Who sees each of the four pictures below? Look for yourself, then match.



4. Who sees each of the pictures below? Draw lines to match.



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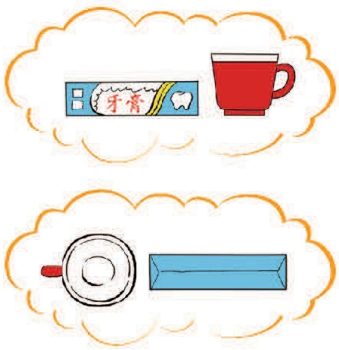
Look (2)

Who sees each of the two pictures below? Think and look.

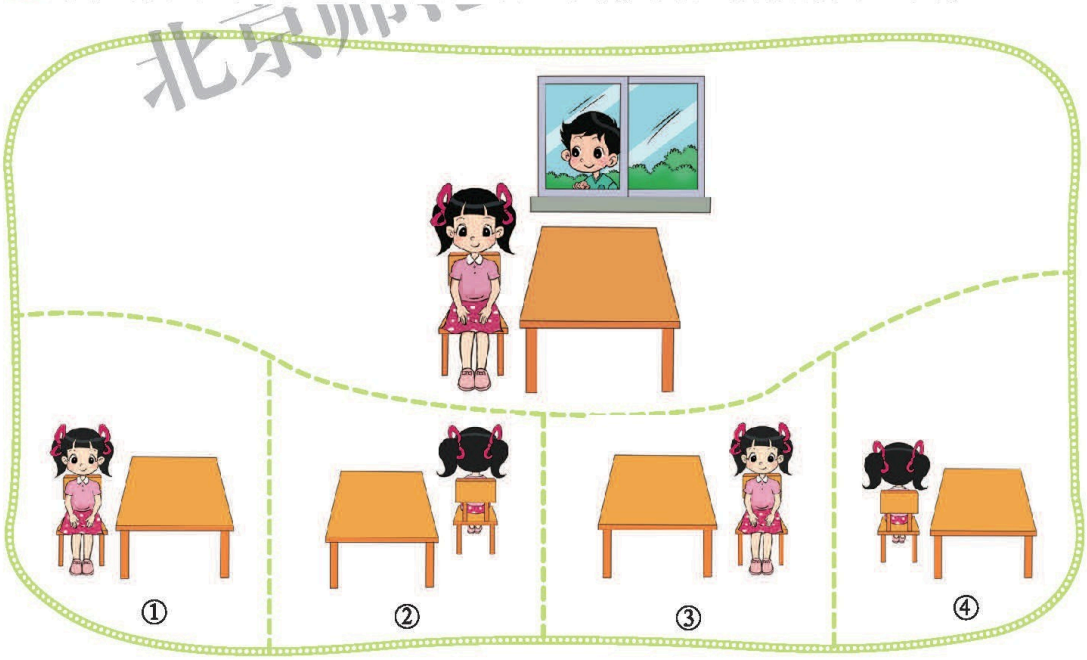


Xiaoxiao says she cannot see the toothpaste box. Is she right?

The cup hides the whole toothpaste box...

Can it really hide it? Let me set it up and see.

Which picture below shows what Taoqi sees through the window? Explain why.



Work with a partner. Act out the scene above to check your choice.

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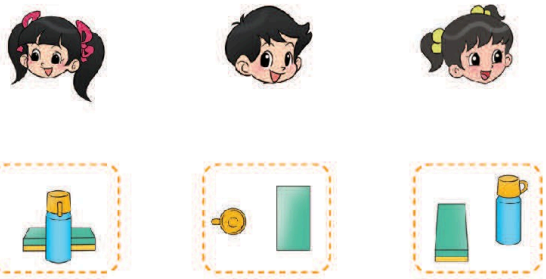
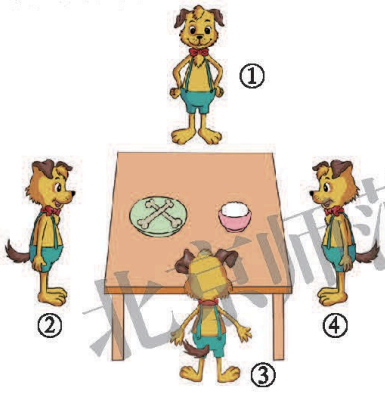
Practice

1. Who sees each of the pictures below? Draw lines to match.

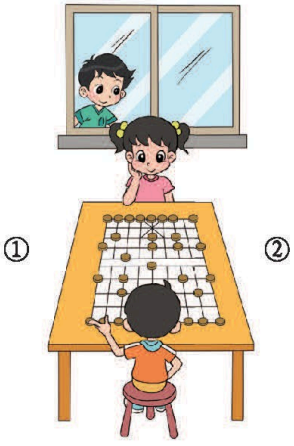


2. From which position does Clever Dog see each picture on the right? Write the number in the brackets.



( ) ( )

3. Look and think

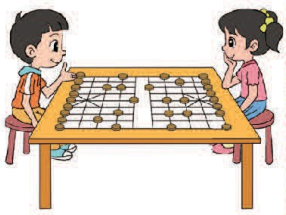


(1) Which picture does Taoqi see? Choose.

( )

( )

(2) Xiaoxiao sees this picture. Which numbered position is she looking from?



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Unit 3 Addition and Subtraction

Book Drive

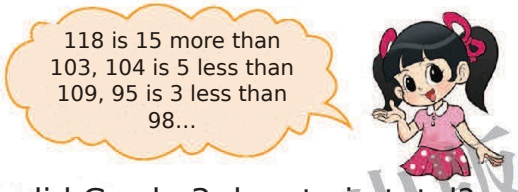
Grade 3 Book Donations

Class	Books
3(1)	118
3(2)	104
3(3)	95

Grade 4 Book Donations

Class	Books
4(1)	103
4(2)	109
4(3)	98

Which grade donated more books? Explain your thinking.



How many books did Grade 3 donate in total?

□ ○ □ ○ □ = □ ( )

$100+100+100=300$

$18+4-5=17$

$300+17=317$

$118+104+95$

$=222+95$

$=317$

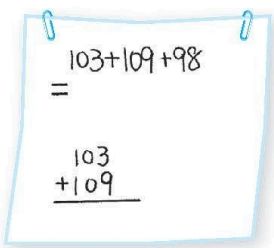
$$\begin{array}{r} 118 \\ +\ 10_{1}4 \\ \hline 222 \end{array} \qquad \begin{array}{r} 222 \\ +\ {}_{1}95 \\ \hline 317 \end{array}$$

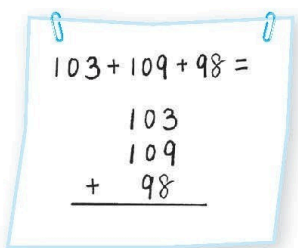
$$\begin{array}{r} 118 \\ 104 \\ +\ {}_{1}9_{1}5 \\ \hline 317 \end{array}$$

When adding three numbers, be extra careful!

Answer: _____ .

How many books did Grade 4 donate in total?





Which grade donated more, and by how many books?

Answer: _____ .

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Practice

1.

Spend ¥1000, get a gift pack

¥140 ¥295 ¥408 ¥320 ¥65

Aunt Zhang: I'll buy a rice cooker, a cassette player and a watch.

Uncle Li: I'll buy a bicycle, a cassette player and a desk lamp.

(1) Who spent more money? Explain your thinking.

(2) How much did Aunt Zhang spend? What about Uncle Li?

(3) Buy one of each item. Which three items get the gift pack? Calculate.

2. Use the column method.

$727+114+86$ $397+304+120$

$475+260+149$ $292+450+108$

3.

50 per bag 100 per bag 150 per bag

(1) How could you buy 200 sweets? (Give two ways.)

(2) If you buy 2 bags of each kind, is that enough for 500 sweets?

4.

Which two numbers add up closest to 500?

220 160 256 290

Which three numbers add up closest to 800?

196 210 385 104

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Cabbage Trucks

After two truckloads are taken away, how many cabbages are left?

First work out how many are left after the first truckload...

$850-256-280$
$=\square-280$
$=\square\ (\ )$

$\begin{array}{r} 850 \\ -\ 256 \\ \hline \square\square\square \end{array}$ $\begin{array}{r} \square\square\square \\ -\ 280 \\ \hline \square\square\square \end{array}$

First find how many both trucks took...

$850-(256+280)$
$=850-\square$
$=\square\ (\ )$

$\begin{array}{r} 256 \\ +\ 280 \\ \hline \square\square\square \end{array}$ $\begin{array}{r} 850 \\ -\ \square\square\square \\ \hline \square\square\square \end{array}$

Answer: _____ .

Can you help them correct the mistakes below?

$\begin{array}{r} 850 \\ -\ 265 \\ \hline 585 \end{array}$ $\begin{array}{r} 585 \\ -\ 280 \\ \hline 305 \end{array}$

Be careful when you calculate...

$\begin{array}{r} 256 \\ +\ 280 \\ \hline 536 \end{array}$ $\begin{array}{r} 850 \\ -\ 536 \\ \hline 326 \end{array}$

After you get the answer, check it.

Calculate, then check your work.

$1000-372-210=\square$

Try thinking of 1000 as 999+1!

$\begin{array}{r} 1\ 0\ 0\ 0 \\ -\ \ 3\ 7\ 2 \\ \hline \square\square\square \end{array}$ $\begin{array}{r} \square\square\square \\ -\ \square\square\square \\ \hline \square\square\square \end{array}$

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Practice

1.

¥78

¥52

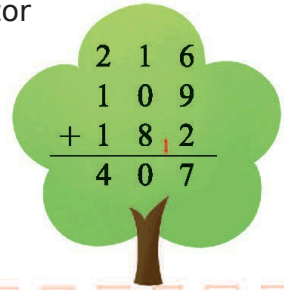
¥35

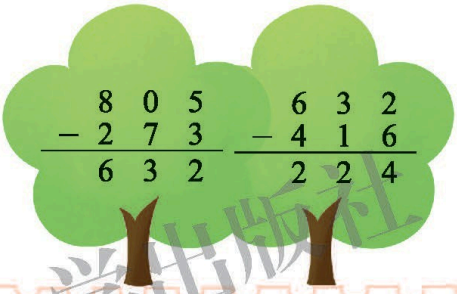
¥75

(1) Aunt Li bought a jacket and a hat. She paid ¥200. How much change should she get?

(2) Make up another problem that uses two subtractions, and solve it.

2. Forest Doctor





3.

$600-233$	$456-278-132$	$368-(168+122)$
$346+199$	$930-(80+480)$	$1000-185-145$

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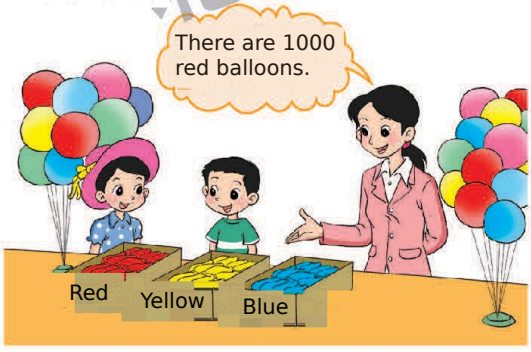
4. Selling balloons



(1) 395 red balloons were sold in the morning and 315 in the afternoon. How many are left?

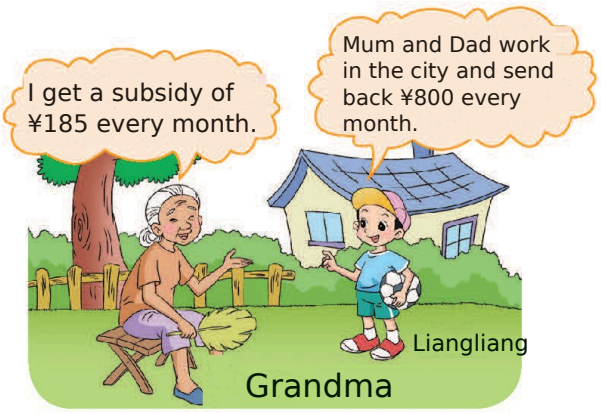
(2) 255 blue balloons were sold in the morning and 340 in the afternoon, leaving 205. How many were there at first?

5. Teacher Zhang accidentally smudged the table below. Can you fill it in? Discuss your method with a partner.

Type	Literature	Science	Biography	Total
At first	315		190	785
Lent out	105			
Left			110	482

Money Saved

In 2009, about 230 million rural migrant workers were working in China's cities. Building and developing cities depends on their hard work.



Liangliang and Grandma spent ¥745 in August. How much did they save?

You can first work out the total income each month, then...

$800+185-745$

$=\square\bigcirc\square$

$=\square$ (yuan)

Or first subtract the money spent from the money Mum and Dad send...

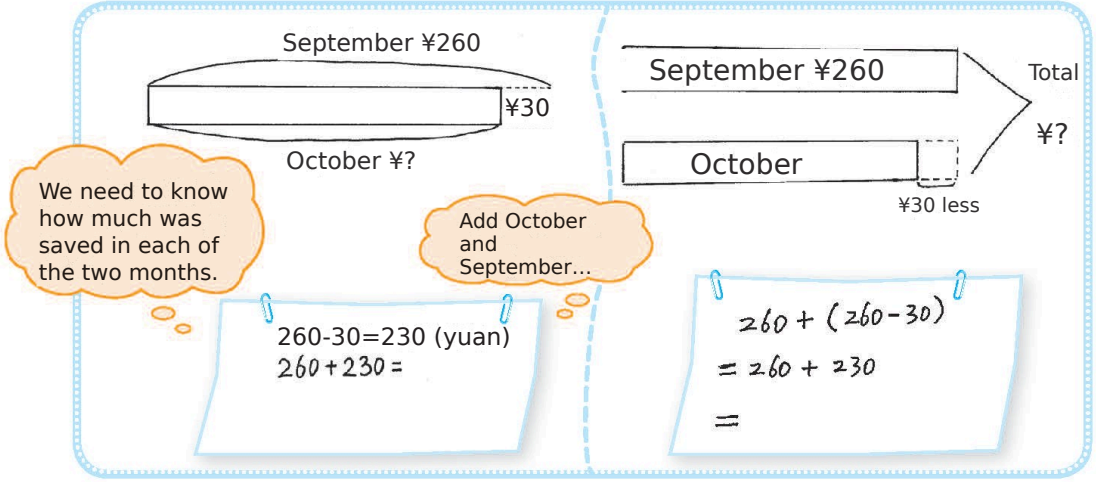
$800-745+185$

$=\square\bigcirc\square$

$=\square$ (yuan)

Answer: _____ .

They saved ¥260 in September. In October they saved ¥30 less than in September. How much did they save in the two months altogether? Explain your thinking.



Answer: _____ .

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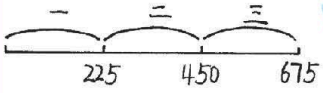
Try It

The family saves ¥225 a month on average. Estimate: how many months of savings will buy a washing machine?

Calculate: how many months must they save to buy a washing machine?



Answer: It takes 3 months.

225+225=450 (yuan)

450+225=675 (yuan)

¥675 > ¥580

Answer: 3 months is enough.

580-225=355 (yuan)

355-225=130 (yuan)

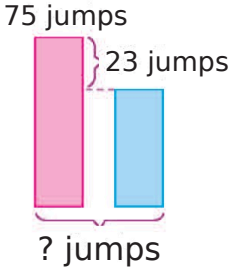
¥130 < ¥225

Answer: Three months' savings can buy a washing machine.

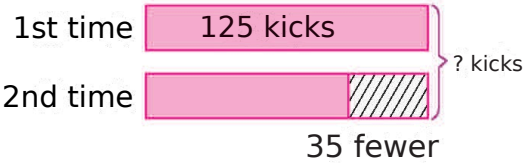
Practice

1. Talk about it, then write a number sentence and solve.

(1) Taoqi jumped rope twice. How many jumps in all?



(2) Xiaoxiao kicked a shuttlecock twice. How many kicks in all?



2. The People's Theatre has 425 seats downstairs and 185 fewer seats upstairs. How many seats does it have altogether? Draw a diagram to explain your thinking, then write a number sentence and calculate.

3.

$278-199+680$	$800-289+484$	$500-294-106$
$318+254-190$	$653+97+203$	$427+273-165$

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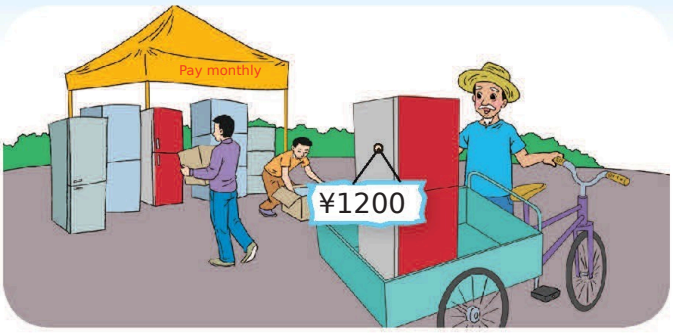
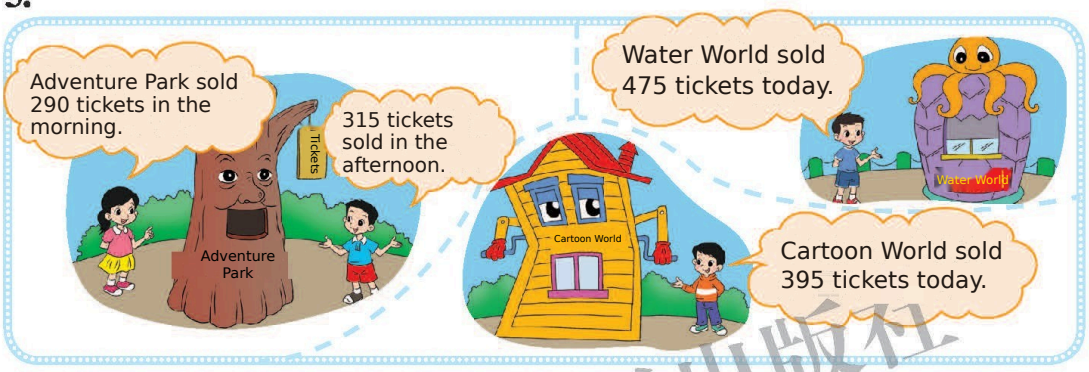
4.



Grandpa Wang bought a refrigerator. He paid ¥500 first and will pay ¥260 a month for the rest. How many months will it take to pay it all off?

5.



(1) How many fewer tickets did Water World sell than Adventure Park?

(2) How many tickets did Adventure Park and Cartoon World sell in total?

(3) Please ask another maths question and try to answer it.

6. Leonardo da Vinci was a famous Renaissance painter and scientist. He left many great paintings and scientific discoveries. In what year was he born? Colour the calculations with answers between 400 and 500 to find out.

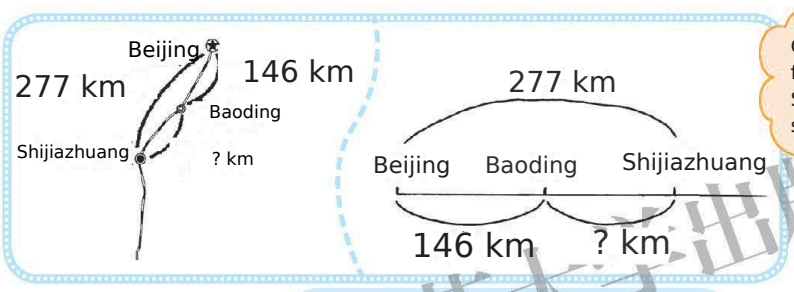
$920-320-150$	$210+120+240$ / $70+360$ / $875-470$ / $649-400$	$420+58$	$235+65+180$	$700-265$	$300+200-65$	$500+135-200$
				$130+460$	$420+580$	$748+22$
		$100+349$		$1000-567$		
$195+165+105$	$392+70$		$650+130$	$900-701$	$690+210-450$	$980-500-20$
	$801-690$	$700-280$			$28+350+50$	$430+190$
	$92+508$		$878-408$			$682-382$
						$830-620+240$

Distance Table (1)

Below is the train distance table for the main stations from Beijing to Xi'an.

	Distance/km
Beijing–Baoding	146
Beijing–Shijiazhuang	277
Beijing–Zhengzhou	689
Beijing–Luoyang	813
Beijing–Xi'an	1200

How far is it from Baoding to Shijiazhuang? Draw a diagram and explain your thinking.



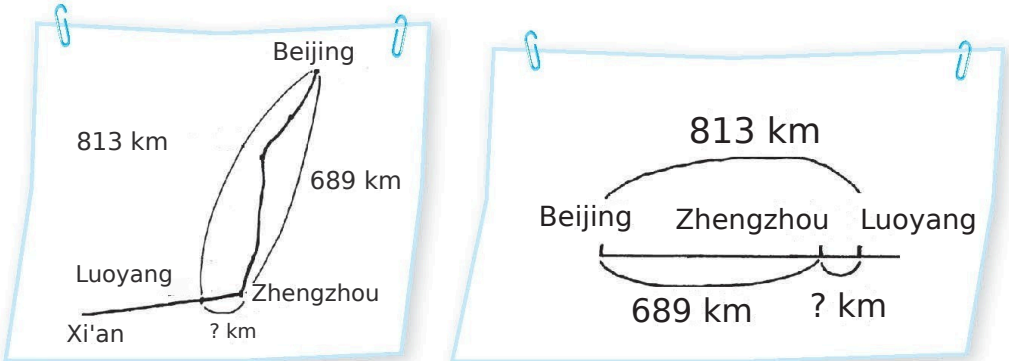
Answer: _____ .

How many kilometres is it from Baoding to Zhengzhou? Draw on the diagram above, then calculate.

□ ○ □ = □ ( )

Answer: _____ .

813 − 689 gives the distance between which two cities? Draw and explain.



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Practice

1. Below is a diagram of the Yangtze River route from Wuhan to Nanjing.

	Distance/km
Wuhan–Jiujiang	269
Wuhan–Wuhu	637
Wuhan–Nanjing	733

(1) How far is it from Jiujiang to Wuhu? Draw and calculate.

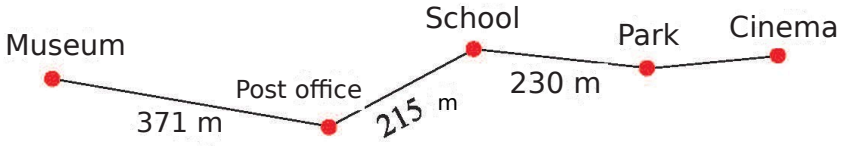
(2) 733 − 637 gives the distance between which two cities? Draw it and explain.

(3) A passenger ship set out from Wuhan and has sailed about 600 km along the route. Mark the ship's approximate position on the diagram.

2.

$653 + 285$	$165 + 435 + 377$	$700 - 399 + 125$
$800 - 672$	$500 - 196 - 204$	$652 - (97 + 355)$

3.



(1) Fill in the table.

	Distance/m
Museum–Post office	
Museum–School	
Museum–Park	

(2) The museum is 990 m from the cinema. How many metres is the park from the cinema?

(3) Which is longer, from the school to the museum or to the cinema? How many metres longer?

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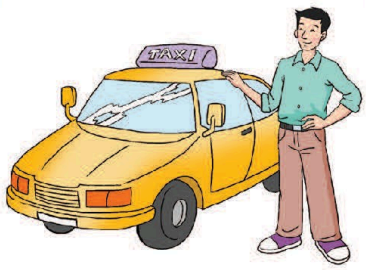
Distance Table (2)

Taoqi's uncle is a taxi driver. When he set out on Monday morning, the odometer read 35 km. Taoqi recorded the odometer reading when his uncle came home each day from Monday to Friday. (Unit: km)



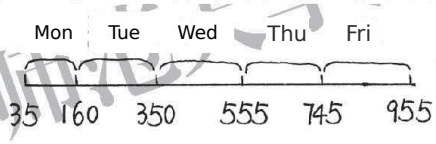
Mon	Tue	Wed	Thu	Fri
160	350	555	745	955

Think, then say what maths information you have found.

On Monday he drove 160 km, the least...

160 km is not the distance he drove on Monday. It should be...

Taoqi drew this diagram. Do you understand it? Discuss with a partner.



The reading was 35 km when he set out on Monday and 160 km when he got home. So 160-35 is the distance he drove on Monday.

I see! Each day's reading minus the day before's reading is that day's distance.

On which day did he drive the farthest?

160-35=125 (km)
350-160=190 (km)
555-350=205 (km)
745-555=

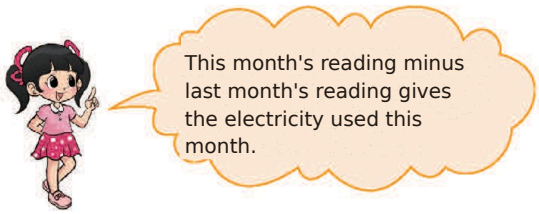
555-350=205 (km)
955-745=210 (km)

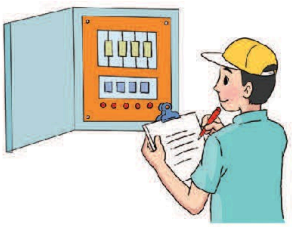
First I estimated: on Wednesday and Friday he drove more than 200 km, then...

Practice

1. Here are the electricity meter readings at Lele's home. (Unit: kWh)

End of Jun	End of Jul	End of Aug	End of Sep	End of Oct	End of Nov	End of Dec
430	545	640	730	825	920	998





(1) What mathematical information have you found? Draw it and talk about it with a partner.

(2) Work out how much electricity Lele's family used in each month from July to December.

(3) How much electricity did Lele's family use in total from July to December (months 7–12)?

(4) Please ask another maths question and try to answer it.

2.

$950-902$	$328+396+204$	$468+376-168$
$567+433$	$900-281-430$	$859-(162+88)$

3. Taoqi's family drove to the ancient capital for a trip. Their home is 1000 km from the ancient capital. They drove 255 km on the first day, 240 km on the second day, and 305 km on the third day.

(1) How many kilometres did they drive in the three days?

(2) How many more kilometres to reach the ancient capital?

4. Uncle Zhao rides a motorcycle to and from work every day. Here is how far he rode each day this week.

	Mon	Tue	Wed	Thu	Fri
Distance/km	28	27	31	29	28

(1) How many kilometres did Uncle Zhao ride in the five days?

(2) When he set off on Monday morning, the odometer read 632 km. Work out what it read when Uncle Zhao got home on Friday evening.

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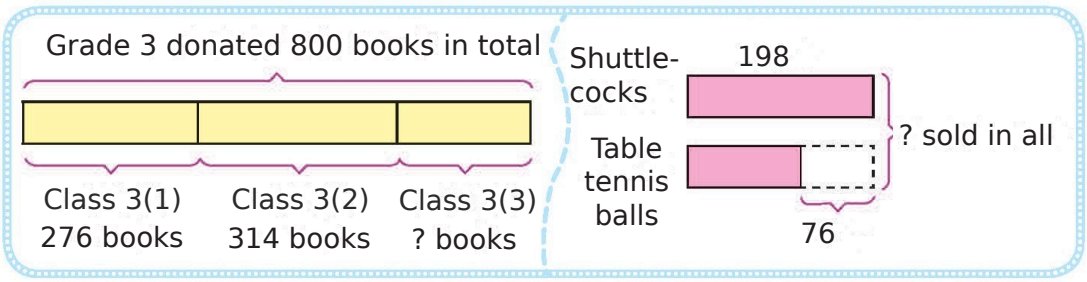
Exercise 2

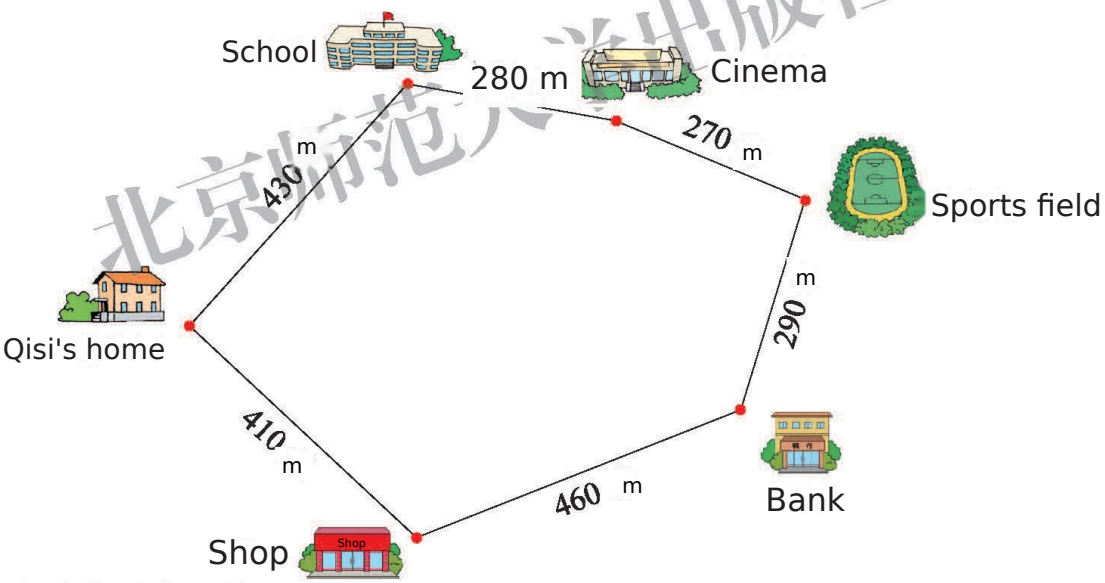
1. Discuss and calculate.



2. Use the column method.

$532+168$	$251+369+142$	$400-245+517$
$962-804$	$738-560-120$	$174+726-378$

3. Which way is shorter from Qisi's home to the sports field? By how many metres?



4. Calculate and fill in.

Grade 1 has 17 fewer pupils than Grade 2.

Grade 1	Grade 2	Grade 3	Total
	216 pupils		600 pupils

5. In the morning, 256 people came to the ski resort. At noon, 142 people left and another 150 arrived. How many people are at the ski resort now?

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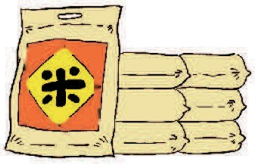
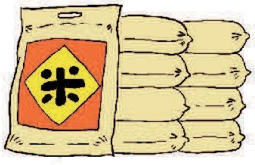
6. How many trips to move all the bags?



280 bags



310 bags

7. Below is the distance table for the main stations on the Qinghai–Tibet Railway, Golmud to Lhasa.

	Distance/km
Golmud–Amdo	694
Golmud–Nagqu	820
Golmud–Damxung	978
Golmud–Lhasa	1142

(1) Draw on the map, then fill in the table.

	Distance/km
Amdo–Nagqu	
Nagqu–Damxung	
Damxung–Lhasa	

(2) 1142 − 820 shows the distance between which two stations? Draw and explain.

(3) Please ask another maths question and try to answer it.

8. □ What digit goes in each box?

$$\begin{array}{r} 7\square 5 \\ +\ \square 7\square \\ \hline 1008 \end{array} \qquad \begin{array}{r} 8\square 0 \\ -\ \square 2\square \\ \hline 64 \end{array} \qquad \begin{array}{r} 6\square\square \\ -\ 425 \\ \hline 1\square 6 \end{array}$$

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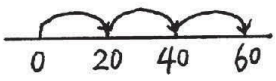
Unit 4 Multiplication and Division

How Many Saplings?

How many saplings are there in 3 bundles? Explain your thinking.

□○□=□ ( )



20 + 20 + 20 = 60

1 ⟶ 20

2 ⟶ 40

3 ⟶ 60

2×3=6

20×3=60

How many in 4 bundles? And in 5?

Answer: _____ .

The gardeners took away 3 truckloads of saplings. How many saplings did they take away in total?

□○□=□ ( )

500 + 500 + 500 = 1500

5×3=15

500×3=1500

Answer: _____ .

Calculate. What do you notice?

8×4=	6×7=
80×4=	6×70=
800×4=	6×700=

Can you write another set of number sentences like these?

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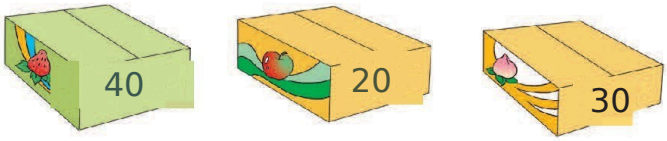
Practice

1. How many of each kind of fruit are there? Calculate and explain your thinking.



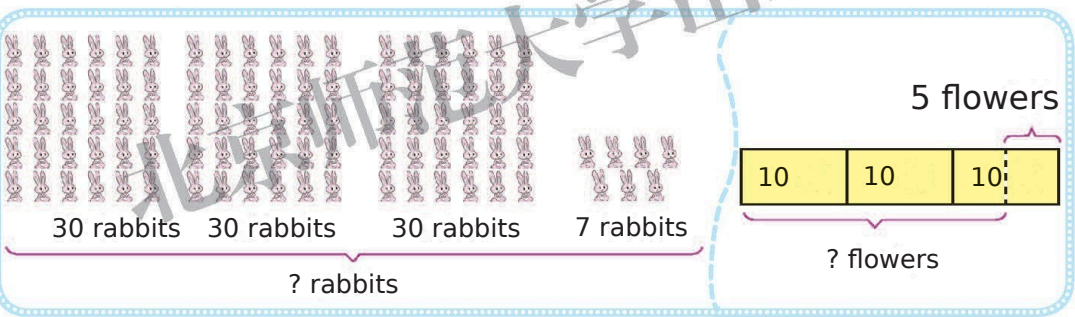
5 boxes　　9 boxes　　7 boxes

2. Calculate and explain your thinking.

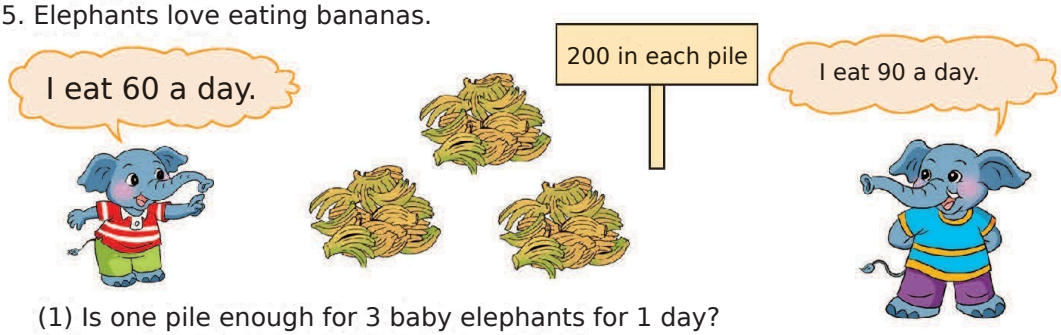
$70 \times 8 =$	$30 \times 6 =$	$600 \times 9 =$
$4 \times 60 =$	$40 \times 5 =$	$5 \times 800 =$

3.

4. Look at the pictures, write number sentences and calculate.



5. Elephants love eating bananas.



(1) Is one pile enough for 3 baby elephants for 1 day?

(2) How many bananas does 1 big elephant eat a week?

(3) One pile of bananas is given to 3 elephants for 1 day, but it is 10 bananas short. Can you work out how many big elephants and how many baby elephants there are?

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How Much Money?

How much do 3 swim rings cost? Explain your thinking.

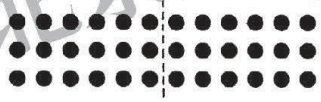
□○□=□ ( )

12+12+12=36

→ 10×3=30
→ 2×3=6
30+6=36

Answer: _____。

Taoqi and Xiaoxiao calculated like this. Do you understand? Talk about it with your partner.



3×6=18　3×6=18

18+18=36

×	10	2
3	30	6

30+6=36

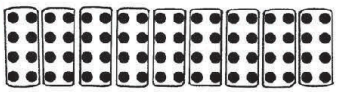
A small boat costs 4 times as much as a ball. How much is the small boat?

□○□=□ ( )

10×4=40
8×4=32
40+32=72

×	10	8
4	40	32

40+32=72

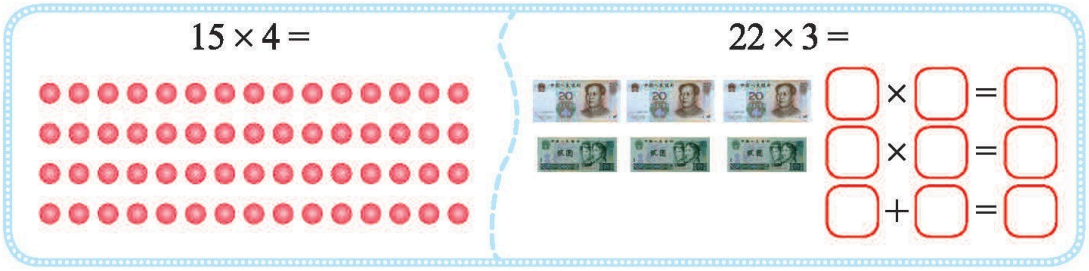


2×4=8
8×9=72

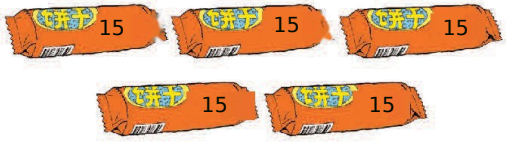
Answer: _____。

Practice

1. Circle and calculate.



2. How many biscuits are there in total? Explain your thinking.



3. Calculate and explain your thinking.

13 × 3 =	12 × 5 =	24 × 2 =	15 × 3 =
31 × 3 =	34 × 2 =	24 × 4 =	13 × 5 =

4. Think and fill in.

14 × 7

×	10	4
7		

□ + □ = □

13 × 6

×	10	3
6		

□ + □ = □

12 × 8

×		

□ + □ = □

5.

Lilies
¥14 each

Roses
¥5 each

Chrysanthemums
¥6 each

Carnations
¥4 each

(1) How much do 6 lilies cost in total?

(2) Aunt Li bought 1 lily and some roses and spent ¥54 in total. How many roses did she buy?

(3) It is Xiaoxiao's grandma's birthday. Mum chose 1 lily and 13 carnations for her. How much do these flowers cost in total?

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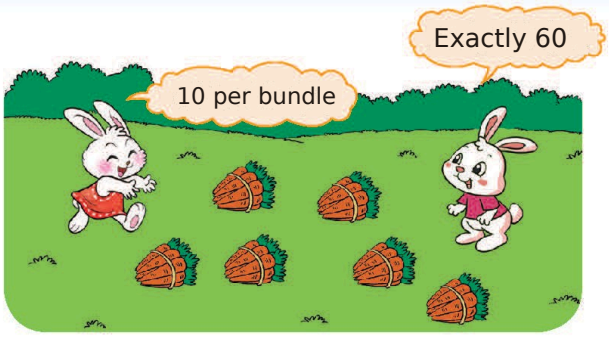
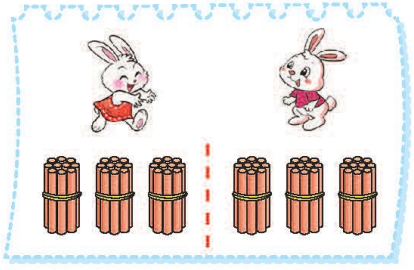
Harvest

How many carrots does each rabbit get on average?

□ ○ □ = □ ( )





$30 \times 2 = 60$

$60 \div 2 = 30$

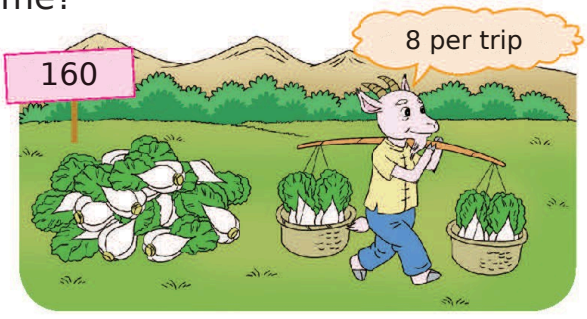
$6 \div 2 = 3$

$6\,0 \div 2 = 3\,0$

Answer: _____ .

How many trips does Uncle Goat need to carry these cabbages home?

□ ○ □ = □ ( )



Answer: _____ .

Calculate. What do you notice?

$42 \div 7 =$

$420 \div 7 =$

$4200 \div 7 =$

$35 \div 5 =$

$350 \div 5 =$

$3500 \div 5 =$

Can you write another set of number sentences like these?

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Practice

1.

80 sweets in total.

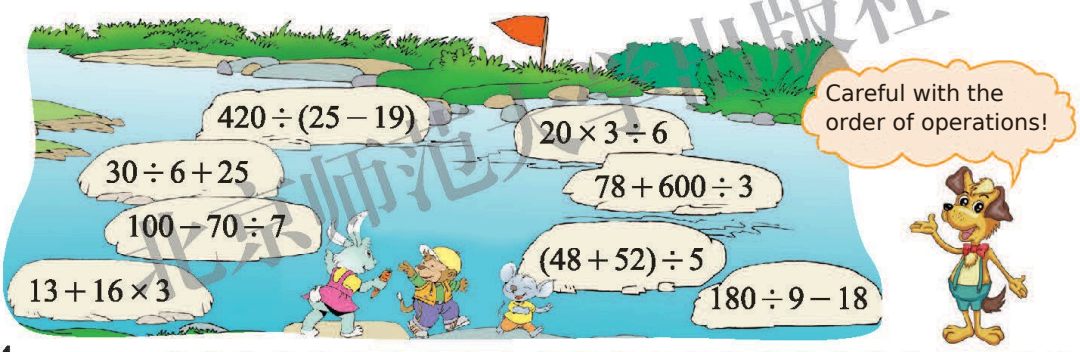
People	Sweets each person gets
2	$80 \div 2 =$
4	
8	

What do you notice?

2. Calculate and explain your thinking.

$60 \div 3$	$200 \div 5$	$240 \div 8$	$120 \div 4$
$560 \div 7$	$900 \div 3$	$210 \div 3$	$270 \div 9$

3. Capture the flag



4.

(1) Qisi read it in 6 days. How many pages a day on average?

(2) Xiaoxiao has read 80 pages and needs to finish the rest in 5 days. How many pages must she read a day on average?

(3) Think of another real-life maths problem that can be solved with (180 − 80)÷5. Talk about it with a partner.

5. Fill in.

(1)

×	30	40	80
2			
6			
7			

(2)

×	20		
3	60		150
5		200	
8			

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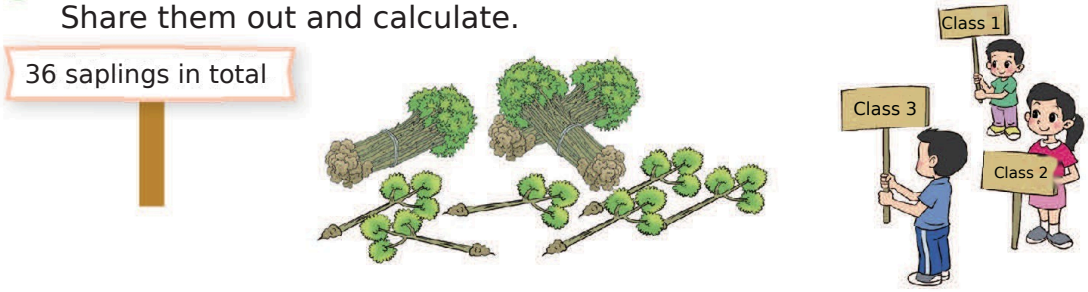
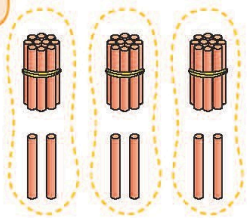
Planting Trees

How many saplings does each class get on average? Share them out and calculate.



□ ○ □ = □ ( )

First share out 3 bundles, then the 6 left. Each class gets 12 saplings.



$30 \div 3 = 10$

$6 \div 3 = 2$

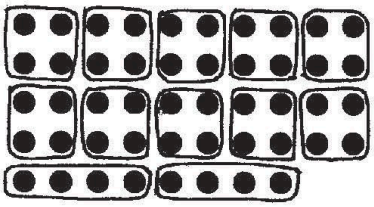
$10 + 2 = 12$

We can show the sharing with number sentences.

Answer: _____。

48 pupils in Class 3 (1) plant trees in groups of 4. How many groups?

□ ○ □ = □ ( )



$40 \div 4 = 10$

$8 \div 4 = 2$

$10 + 2 = 12$

Answer: _____。

Calculate. What do you notice?

$30 \div 3 =$

$33 \div 3 =$

$36 \div 3 =$

$39 \div 3 =$

I can write another set of number sentences:
$40 \div 4 = 10$
$44 \div 4 = 11$
$48 \div 4 = 12$

Can you explain what you found? Use it to work out $52 \div 4$.

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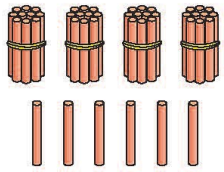
Practice

1. Share out, calculate and explain your thinking.

$46 \div 2 =$

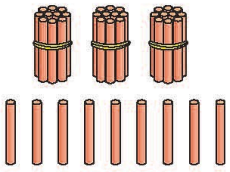


$40 \div 2 = \square$

$\square \div \square = \square$

$\square + \square = \square$

$39 \div 3 =$



$\square \bigcirc \square = \square$

$\square \bigcirc \square = \square$

$\square \bigcirc \square = \square$

2.

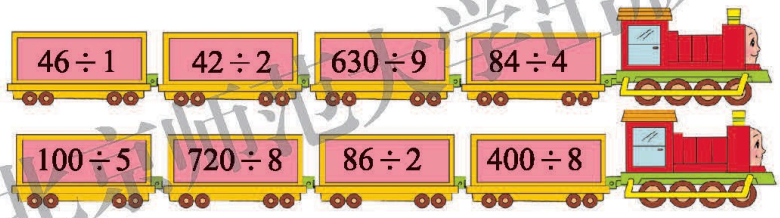
¥5

¥55

(1) The shoes cost how many times as much as the gloves?

(2) How much more do the shoes cost than the gloves?

3. Race.



4. Teacher Wang has ¥100 to buy stationery for the class.

(1) If she buys only pens, how many can she buy?

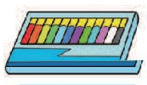
(2) If she buys a box of markers, how many set squares can she buy with the money left?

(3) Teacher Wang bought some identical items with ¥100 and had ¥12 left. What might she have bought?

Set square ¥3

Pen ¥2

Diary ¥4



Markers ¥7

Pencil case ¥6

5. Keep going. How many number sentences can you write?

$60 \div 3 = 20$

$63 \div 3 = 21$

$66 \div 3 = 22$

$69 \div 3 = 23$

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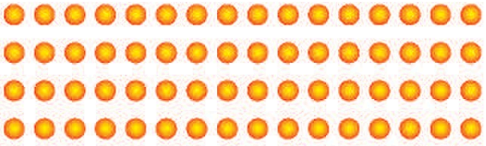
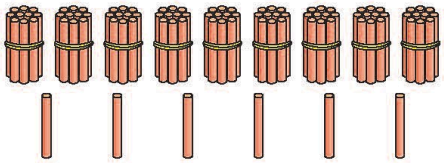
Exercise 3

1. Circle, calculate and discuss with a classmate.

$16 \times 4 =$



$96 \div 3 =$

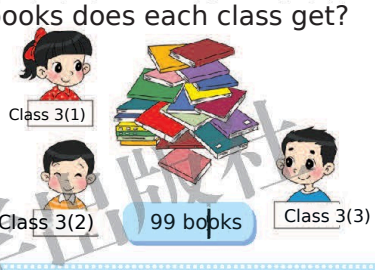


2. Look at each picture, write a number sentence and work it out.

How many stickers are there in total?

24 stickers　24 stickers　24 stickers

If shared equally, how many books does each class get?



3. ☐ What number goes in the box?

☐ × 2 = 60	☐ × 2 = 24	☐ × 3 = 60	☐ × 3 = 12
☐ ÷ 2 = 60	☐ ÷ 2 = 24	☐ ÷ 3 = 60	☐ ÷ 3 = 12

4.

12×4	16×5	$88 \div 4$	$400 \div 5$
23×3	36×2	$900 \div 3$	$62 \div 2$

5. Calculate, then write another set of number sentences like these.

$3 \times 7 =$		$63 \div 9 =$	
$30 \times 7 =$		$630 \div 9 =$	
$300 \times 7 =$		$6300 \div 9 =$	

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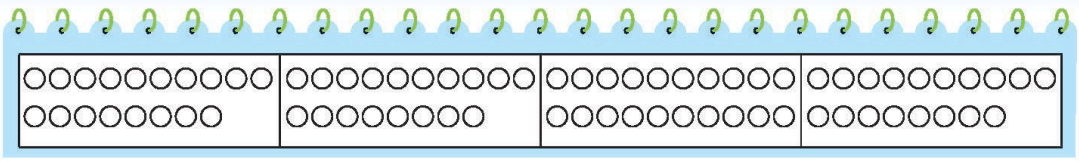
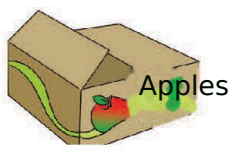
6. The little rabbit recorded how many mushrooms it picked over 4 days.

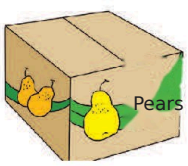


How many mushrooms did the little rabbit pick in total?

7.



¥17 per box



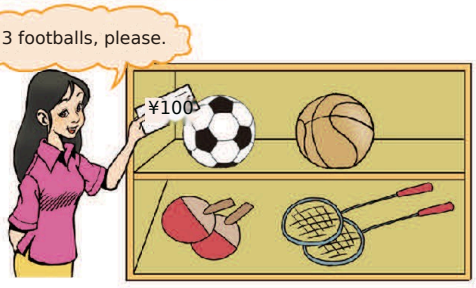
¥15 per box

(1) How much are 4 boxes of apples? How much are 6 boxes of pears?

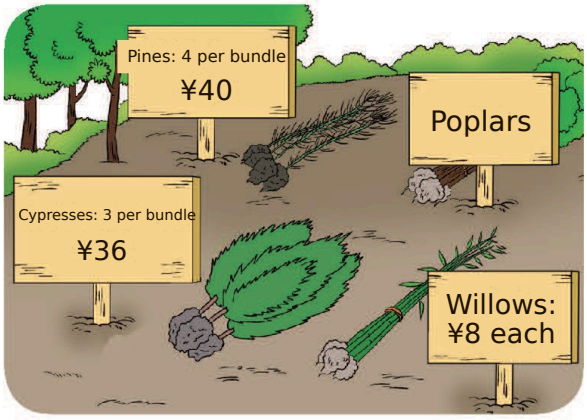
(2) The kindergarten bought 3 boxes of apples and 3 boxes of pears. How much did they cost in total?

8. How old are Xiaodong's father and mother this year?

9. Teacher Li gave the shop assistant ¥100 and got ¥7 change. How much does each football cost?



10.



(1) How much is each cypress?

(2) Compare one pine and one willow. Which costs more, and by how much?

(3) Class 3(1) used the ¥100 donated by the whole class to buy 6 cypresses. How much money was left? The rest was just enough to buy 4 poplars. How much does each poplar cost?

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Review

What I Learned

Look

Who sees each of the three views below? Talk about it.

Work in groups. Choose an object, look at it from different angles and say what you see.

Survey

Xingfu Township Notice Board

Village	Homes with cable TV
Taohua	214
Xinghua	147
Zaohua	153
Lihua	
Total	650

How many homes in the first three villages have cable TV? Does Lihua Village have more than 100?

Share a good way to do multi-digit addition and subtraction correctly, and discuss it with a partner.

Sunshine Sports

23 boys and 25 girls, divided equally into 4 columns

There are 11 groups in total. Only 1 group has 3 people, and each of the other 10 groups has 5 people.

Each group has 8 boys and 8 girls, with 6 groups in total

There are 35 of us in Class 1.

Our Class 2 is divided into 3 groups of 12 students.

Using the picture, say what each number sentence below means. Then work them out and say the order you calculated in.

$(23+25)\div 4$ $12\times 3-35$ $10\times 5+3$ $(8+8)\times 6$

My Progress

My proudest answer in class was when I thought of different ways to solve a problem, and the teacher and my classmates all clapped for me.

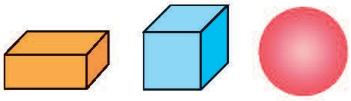
My favourite maths activity was copying the children in the book and looking at objects from different positions.

I like drawing pictures to solve real-life problems.

My Questions

Why does the same object look small from far away and big from close up?

A big cuboid or cube shows at most three faces. Why does this small cube also show only three faces? And a ball?



Can a number sentence have two "( )"? Which do we work out first? What does "[ ]" mean? How do we use it?

$86 \div 3 =$

$628 \div 4 =$

Can all these be worked out with long division? How do we set them out?

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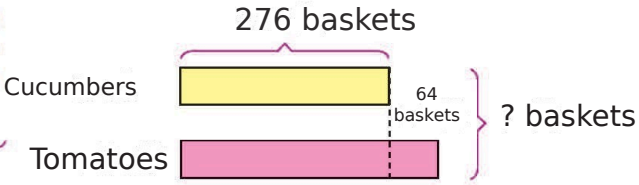
Consolidate and Apply

1. Talk about the pictures, then write a number sentence and work it out.

12 peaches 12 peaches 12 peaches

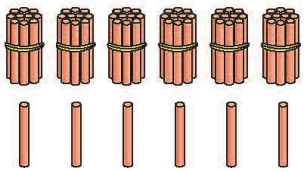
? peaches



2. Circle and fill in.

$14 \times 3 =$

$66 \div 3 =$



$25 \times 2 =$

×	20	
2		

3.

15×6	23×4	$68 \div 2$	$720 \div 9$
16×5	12×7	$99 \div 9$	$350 \div 5$

4.

$45 + 15 \times 3$	$80 \div 4 + 36$	$871 - 381 + 129$
$126 - 96 \div 3$	$5 \times (36 - 19)$	$(83 - 35) \div 4$

5. Forest Doctor

$384 + 136 - 255$
$= 510 - 255$
$= 345$

$7 \times 4 + 15$
$= 28$
$= 43$

$24 + 40 \div 8$
$= 64 \div 8$
$= 8$

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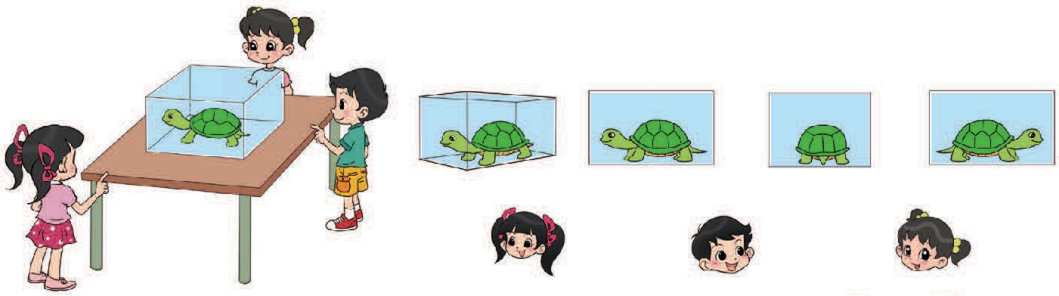
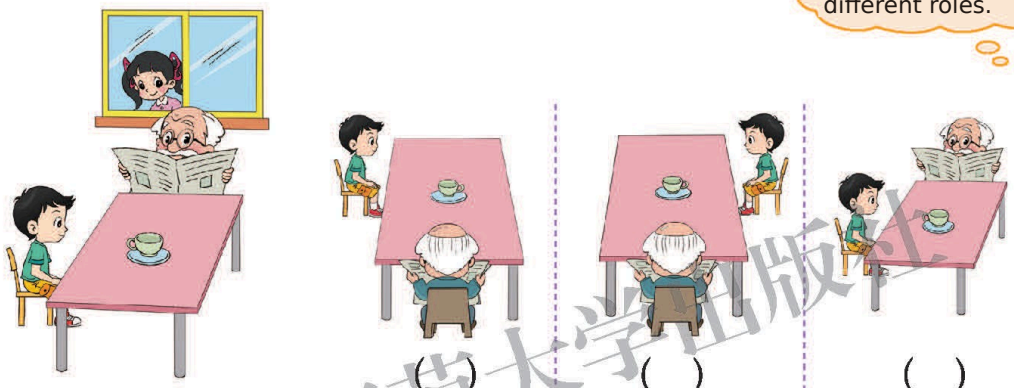
6. Try it out and match.



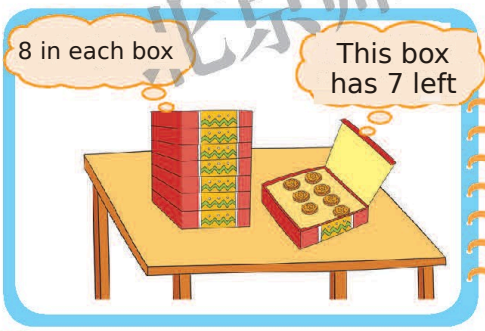
7. Which picture does Xiaoxiao see? Put a ✓ in the ( ).

Act it out with classmates in different roles.



( )　( )　( )

8. Sharing mooncakes at the kindergarten.



(1) How many mooncakes in total?

(2) If the mooncakes are shared equally among the children in 3 classes, how many does each class get?

9.

¥2680

Xiaojun

(1) Xiaojun's family needs ¥2500 a month for living costs. How much can they save each month?

(2) After how many months of saving can they buy this TV?

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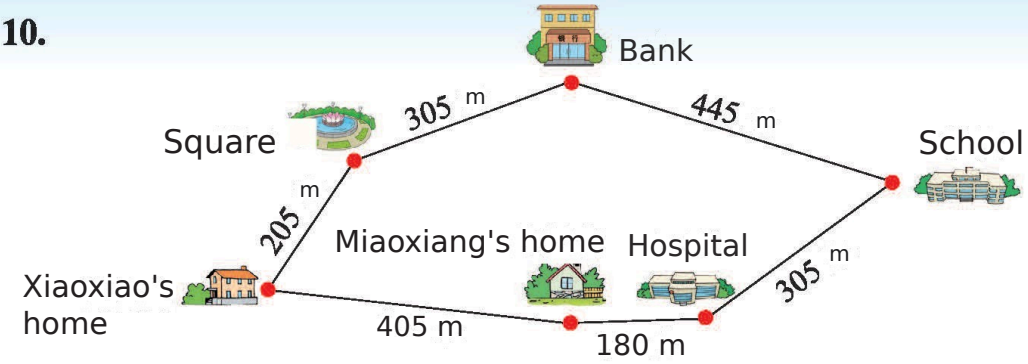
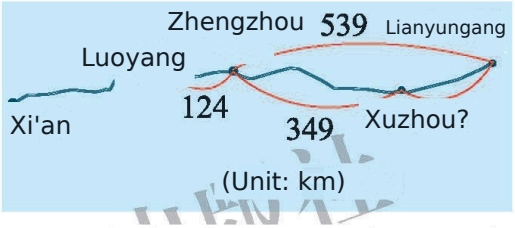
10.



(1) Xiaoxiao has ( ) routes to school. Which is shorter? Mark it on the map.

(2) Miaoxiang wrote 180+305 − 405. What problem was she solving?

11. (1) How far is Xuzhou from Lianyungang?



(2) Zhengzhou to Xi'an is 511 km.
How far is it from Luoyang to Xi'an?

Draw and calculate.

(3) A train from Zhengzhou to Xi'an has gone 400 km. Mark its rough position on the diagram.

12. Below are Uncle Li's car odometer readings at the end of each day, Monday to Friday. (Unit: km)

Mon	Tue	Wed	Thu	Fri
360	525	640	730	905

Monday's starting reading was 187 km.

(1) What information does the table give? Draw it and discuss with a partner.

(2) Work out how far Uncle Li drove each day. On which day did he drive the most? The least?

13.

(1) How many pages a day did Miaoxiang read on average in the first 3 days?

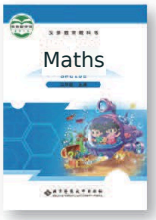
(2) The storybook Miaoxiang borrowed has 119 pages in total. At this rate, can she finish it in a week?

(3) To finish on time, how many pages a day must she read on average from day 4 on?

Unit 5 Perimeter

What Is Perimeter

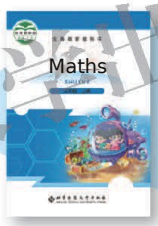
Use coloured pens to trace the edges of the leaf and the maths book cover.

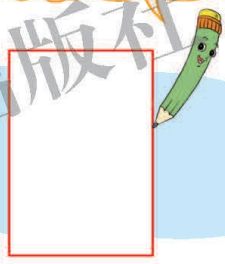


Look and say

The distance I crawl once around the leaf is its perimeter.

The distance around a shape is its perimeter.





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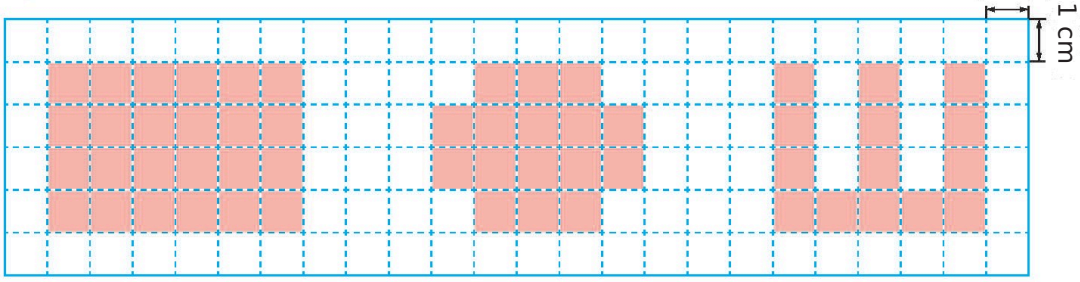
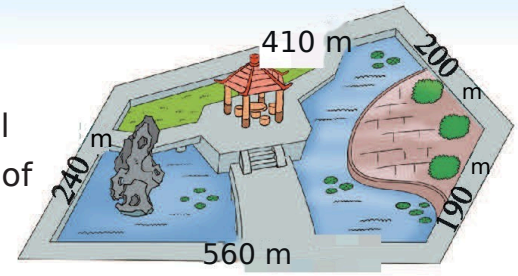
Can you find the perimeter of the leaf and the maths book cover? Work with a partner and measure.

Count. How many centimetres is the perimeter of each shape below?



Try It

A path goes around a small park. What is the perimeter of the small park in metres?



240+410+200+190+560=1600 (m)

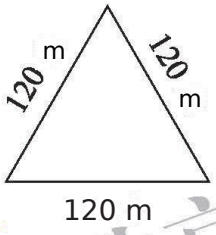
190+410=600 (m)

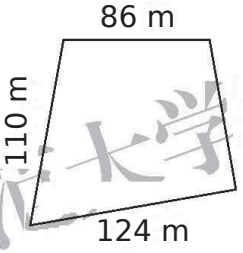
560+240=800 (m)

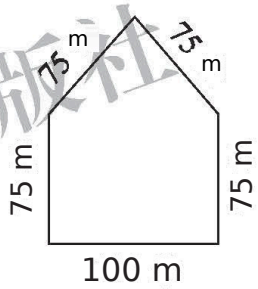
600+800+200=1600 (m)

Answer: _____ .

Find the perimeter of the figures below.







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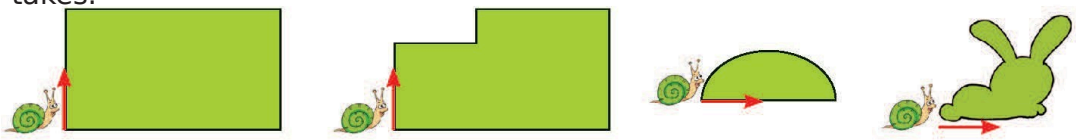
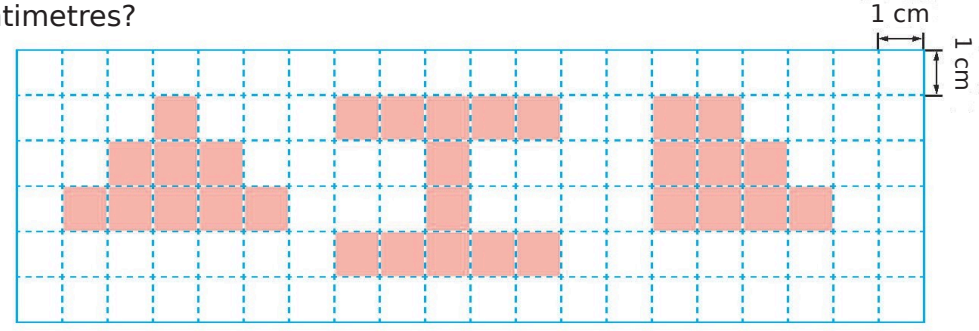
Practice

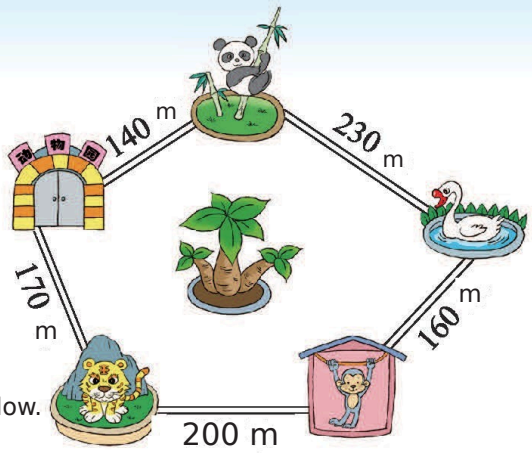
1. Each snail walks once around the edge of its figure. Draw the route it takes.



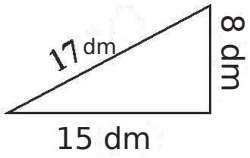
2. Count. What is the perimeter of each figure below in centimetres?

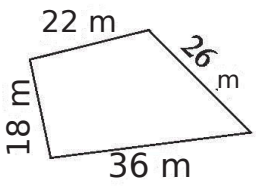


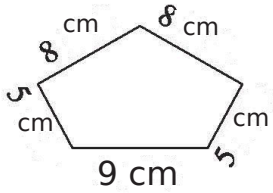
3. In the zoo, Xiaoxiao walked once around the route on the right. How many metres did she walk in total?



4. Calculate the perimeter of the figures below.





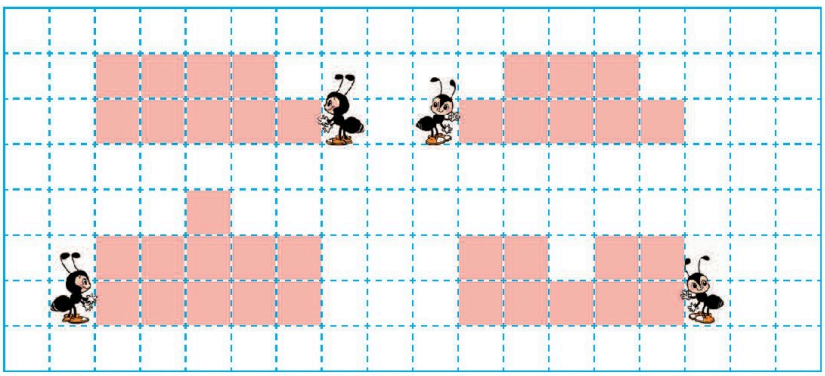


5. Use 4 small squares with 1 cm sides to make 3 different shapes. What is the perimeter of each shape?

6. Estimate and measure.

Perimeter of:	Estimate	Measure
exercise book cover		
desk top		
sole of your shoe		

7. 4 ants each walk once around one of the figures below. Are their routes the same length?



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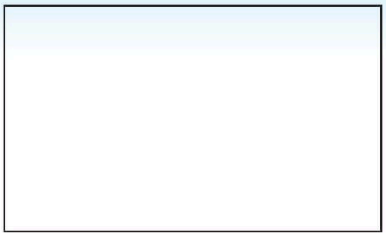
Rectangle Perimeter

Measure the rectangle on the right and work out its perimeter. Explain your thinking.



Measure the lengths of the 4 sides, then add them up.

5+3+5+3=16 (cm)

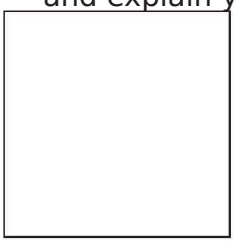
Measure the length and width, then add 2 lengths and 2 widths.

5×2+3×2=16 (cm)

First add 1 length and 1 width, then multiply by 2.

(5+3)×2=16 (cm)

How do you calculate the perimeter of a square? Measure, calculate and explain your thinking.



Add the lengths of the 4 sides. 3+3+3+3=12 (cm)

All 4 sides are 3 cm, so just multiply the side length by 4. 3×4=12 (cm)

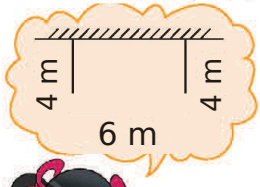
How do you find the perimeter of a rectangle or square?

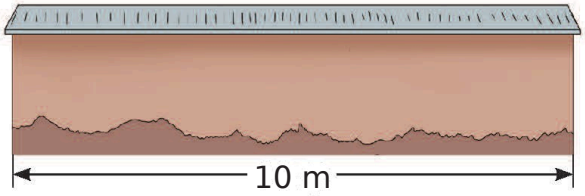
Just add up all their sides.

You can use their properties. For example, the perimeter of a square is the side length times 4.

Taoqi wants to fence a 6 m by 4 m rectangular vegetable patch against a wall.

How can he fence it? How long is the fence each way?

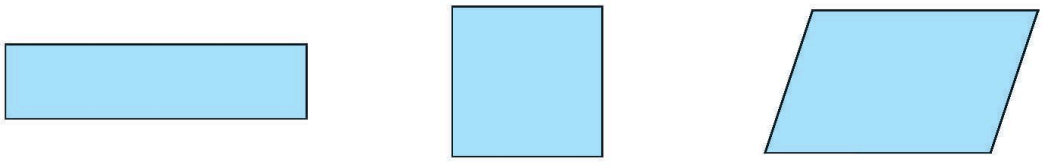




There is another way...

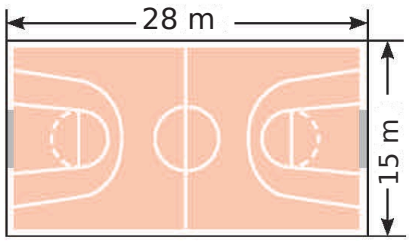
Practice

1. Measure, then calculate the perimeter of each figure.

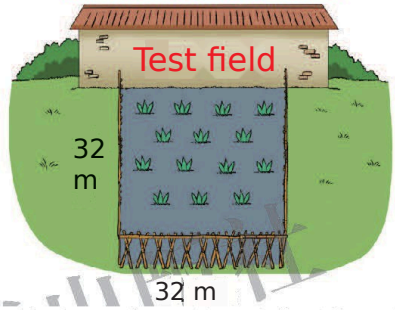


2.

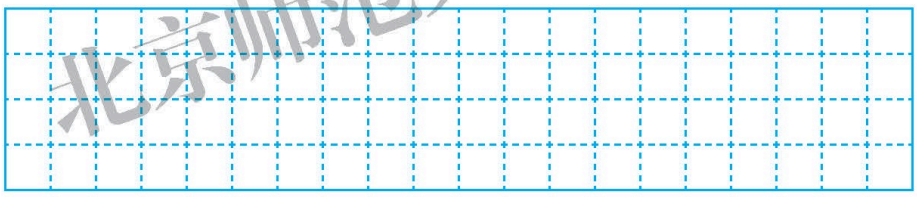
What is the court's perimeter?



How long is the fence?

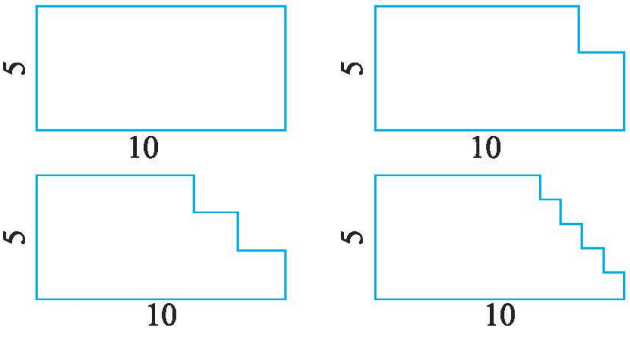


3. On the grid paper below, draw 2 rectangles and 2 squares, and find their perimeters. (Take the side of each small square as 1 cm.)



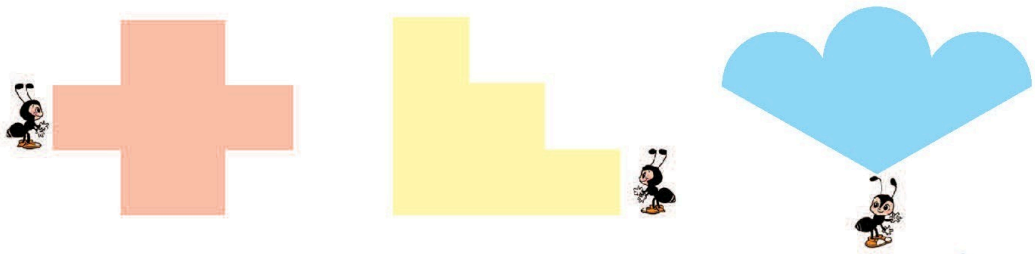
4. Taoqi used a 40 cm piece of string to make a square. What is the side length of the square?

5. Can you find the perimeter of the figures below? (Unit: m)

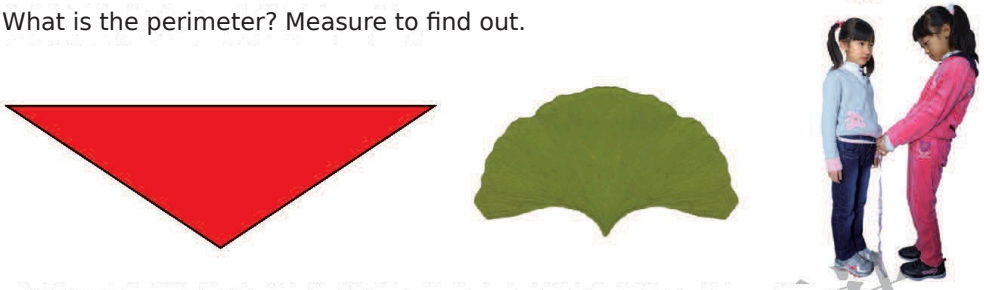


Exercise 4

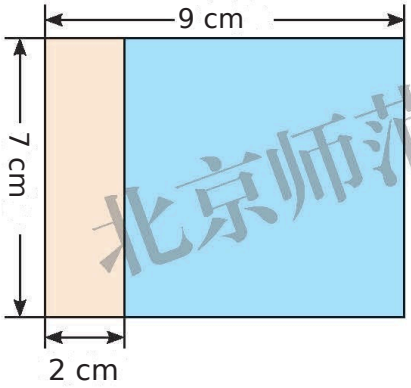
1. Each ant walks once around the edge of its figure. Trace the path it takes.



2. What is the perimeter? Measure to find out.



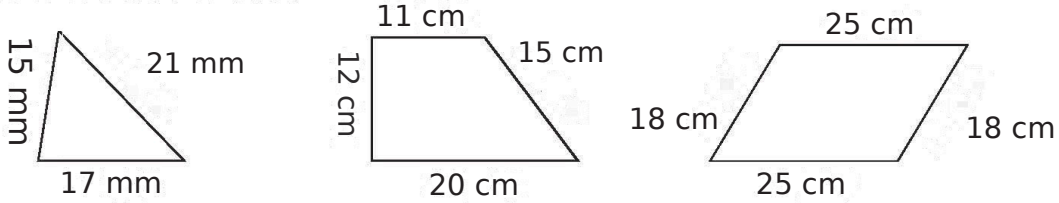
3. Which figure's perimeter does each number sentence on the right find? Explain.



$(7+2)\times 2$

$(9+7)\times 2$

4. Calculate the perimeter of each figure.



5. Calculate each perimeter.

Rectangle

Length	20 m	6 dm	15 cm
Width	10 m	4 dm	6 cm
Perimeter			

Square

Side	7 cm	14 dm
Perimeter		

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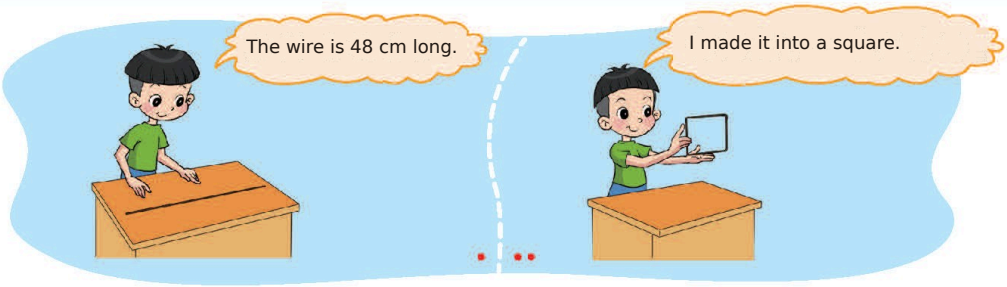
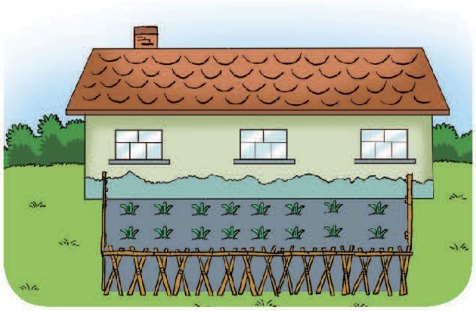
6. Look and work out the side length of the square in centimetres.



7.

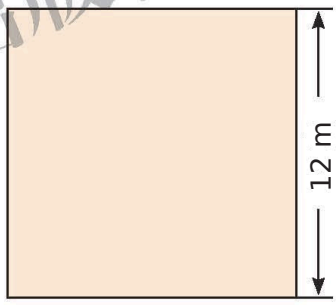


A rectangular vegetable plot is 4 m long and 1 m wide. One side is a wall. How many metres long is the fence around it?

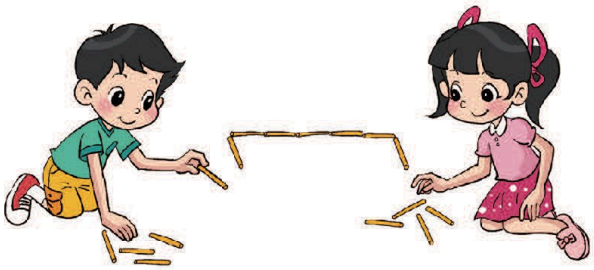
8. (1) Find the perimeter of the large square.

(2) Divide this large square into 4 equal small squares. Draw them and find the perimeter of each small square.

(3) How does the total perimeter of the 4 small squares compare with the perimeter of the large square? Can you explain why?



9. Try it and fill in the table.

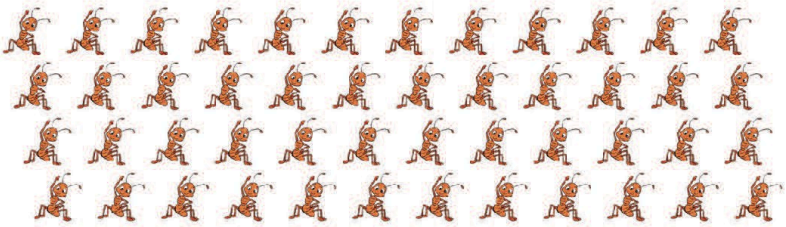


Use 16 sticks of the same length to make different rectangles. How many can you make? How many sticks long are their lengths and widths? Work with your partner.

Length (sticks)				
Width (sticks)				

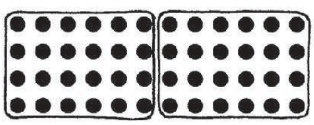
Unit 6 Multiplication

Ant Exercises



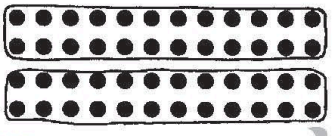
How many ants are there in total? Circle groups on the dot diagram and calculate.

□ ○ □ = □ ( )



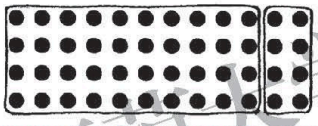
6×4=24

24×2=48



12×2=24

24×2=48



10×4=40

2×4=8

40+8=48

Answer: _____ .

Look and explain what each step of the column method means.

×	10	2
4	40	8

T O

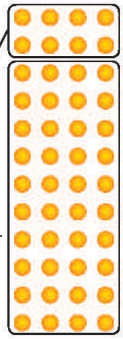
1 2

× 4

8 → 2×4

+ 4 0 → 10×4

4 8



We usually write it like this.

1 2

× 4

4 8

Try to work it out.

213×3=□

200×3=600

10×3=30

3×3=9

600+30+9=639

□□□

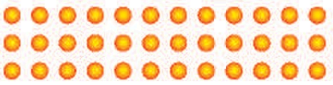
×□

Practice

1. Circle and calculate.

$14 \times 2 = \square$

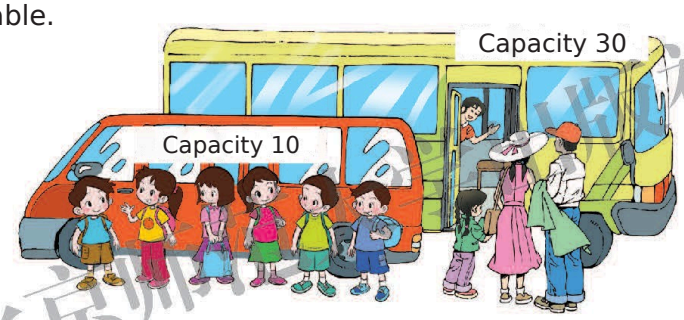
$12 \times 3 = \square$



2. Calculate and explain what each step of the column method means.

$$\begin{array}{r} 22 \\ \times\ \ 4 \\ \hline \end{array} \qquad \begin{array}{r} 31 \\ \times\ \ 3 \\ \hline \end{array} \qquad \begin{array}{r} 123 \\ \times\ \ \ 3 \\ \hline \end{array} \qquad \begin{array}{r} 214 \\ \times\ \ \ 2 \\ \hline \end{array}$$

3. Fill in the table.



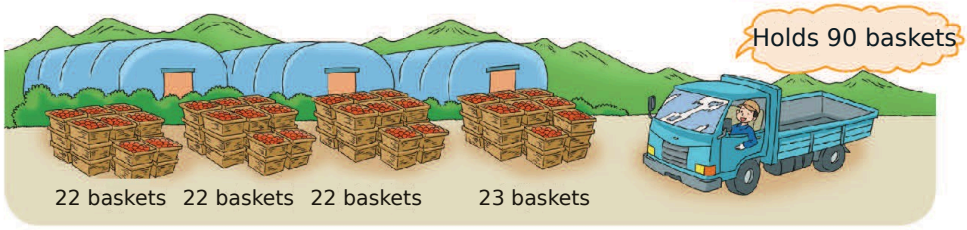
	2 buses	3 buses	4 buses	5 buses	6 buses	7 buses	8 buses
Big bus passengers	60						
Minibus passengers	20						

4. 1 coat costs 2 times as much as 1 sweater.

(1) How much is 1 coat?

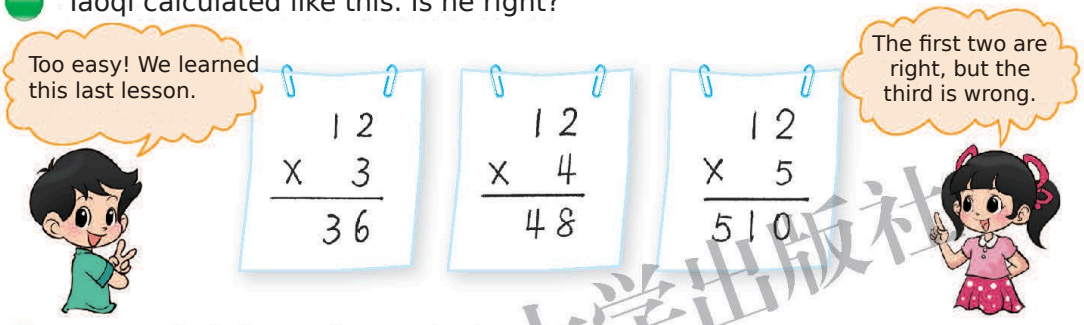
(2) Lele's mum bought her family 2 vests and 1 sweater. How much did she pay in total?

5. Can the lorry take them all in one trip?



Fun Fair

Taoqi calculated like this. Is he right?



How much is it for 5 people to ride the spaceship?

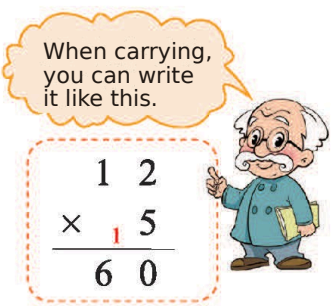
□ ○ □ = □ ( )

$10 \times 5 = 50$
$2 \times 5 = 10$
$50 + 10 = 60$

×	10	2
5	50	10

$50 + 10 = 60$

$$\begin{array}{r} 12 \\ \times\ 5 \\ \hline 10 \\ 50 \\ \hline 60 \end{array}$$



Answer: _____

How much is it for 13 people to ride the electric train?

□ ○ □ = □ ( )

$$\begin{array}{r} 13 \\ \times\ 6 \\ \hline \end{array}$$

What about 14 people? 15 people? 16 people?

Answer: _____

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Practice

1. Do it and calculate.

$16 \times 3 = \square$

$13 \times 7 = \square$

×	10	3
7		

$\square + \square = \square$

2.

2 3	1 6	4 8	1 9
× 2	× 5	× 2	× 2

3. Match.

2×32	15×4	16×5	11×8
5×12	14×7	14×7	4×30
3×16	24×2	22×4	15×6
17×4	4×16	3×40	49×2

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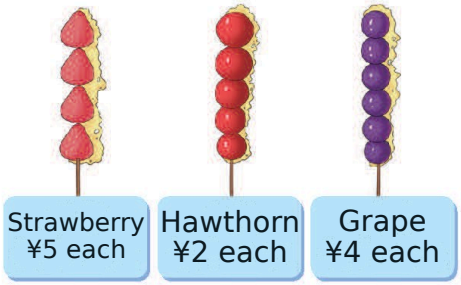
4. Buying candied fruit skewers

(1) Panda Class bought 18 strawberry skewers. How much do they pay?

(2) Sparrow Class bought 16 grape skewers. How much do they pay?



5.

$12 \times 2 =$

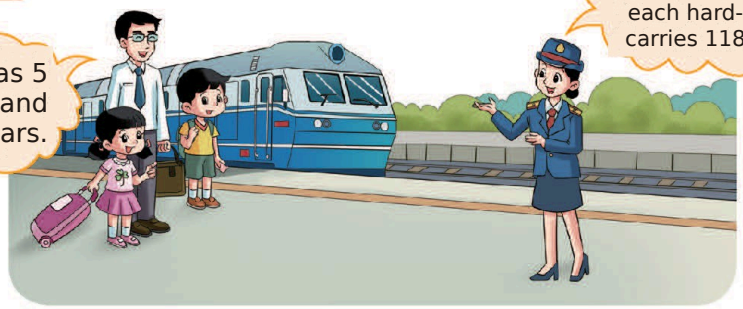
$12 \times 3 =$

$12 \times 4 =$

$12 \times 5 =$

Calculate. How much more is each result than the one before it? Tell a partner why. Then write two more number sentences.

By Train



Talk about it: what maths questions can you ask?

How many people can 5 sleeper cars carry?

□ ○ □ = □ ( )

70 × 5 = 350
2 × 5 = 10
350 + 10 = 360

×	70	2
5	350	10

350 + 10 = 360

$$\begin{array}{r} 72 \\ \times\ 5 \\ \hline 360 \end{array}$$

Answer: _____

How many people can 7 hard-seat cars carry? Calculate and fill in.

□ ○ □ = □ ( )

100 × 7 = 700
10 × 7 = 70
8 × 7 = 56
700 + 70 + 56 = 826

$$\begin{array}{r} 118 \\ \times\ 7 \\ \hline 826 \\ {\scriptstyle 5} \end{array}$$

Don't forget to add the number you carried!

Answer: _____

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Practice

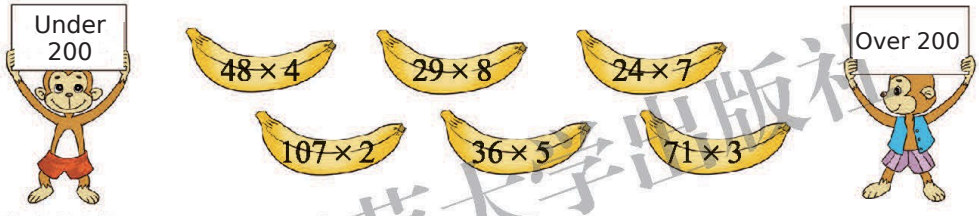
1. Pick mushrooms.

Talk about it: what maths questions can you ask? Try to answer them.

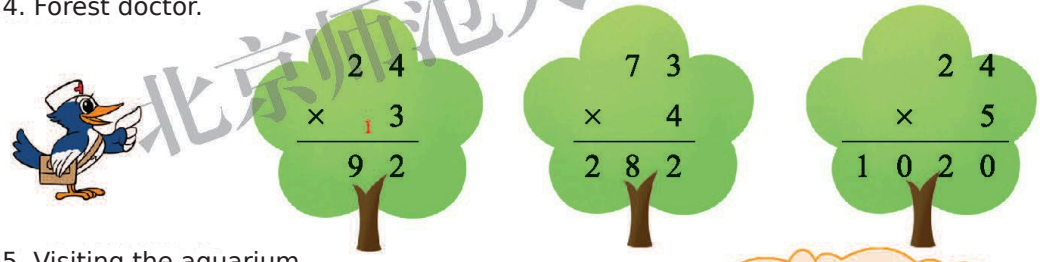
2.

$$\begin{array}{r} 18 \\ \times\ 3 \\ \hline \end{array} \qquad \begin{array}{r} 56 \\ \times\ 8 \\ \hline \end{array} \qquad \begin{array}{r} 125 \\ \times\ 6 \\ \hline \end{array}$$

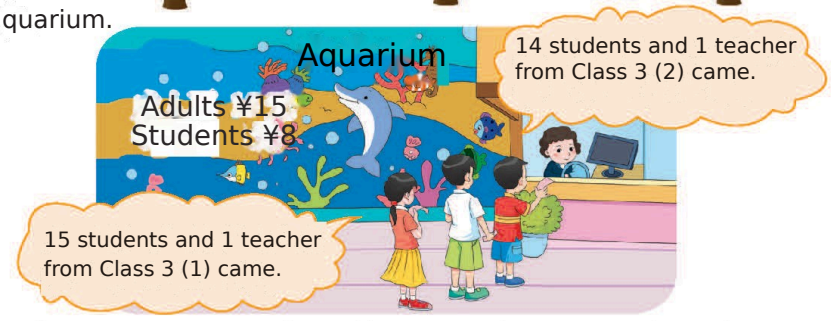
3. Pick bananas.



4. Forest doctor.



5. Visiting the aquarium.



(1) What maths information can you get from the picture? What questions can you ask?

(2) How much does each class need to pay for tickets?

(3) On Children's Day, buy 10 student tickets and get 1 adult ticket free. How much do the two classes pay for tickets in total?

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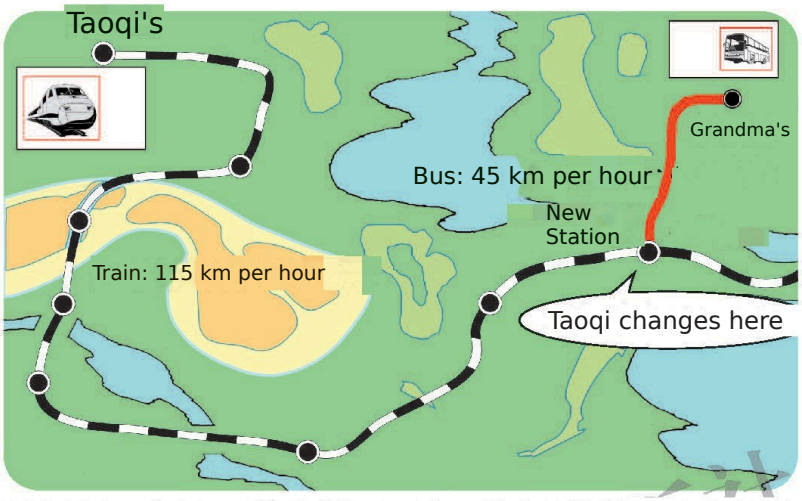
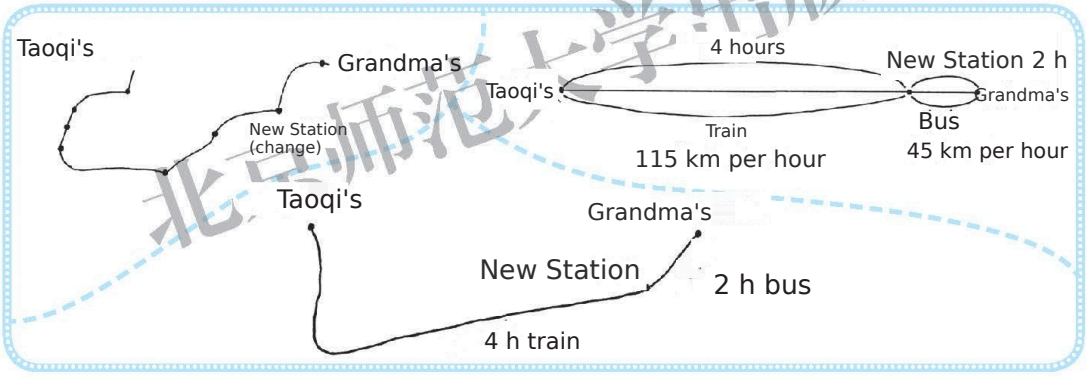
Visiting Grandma

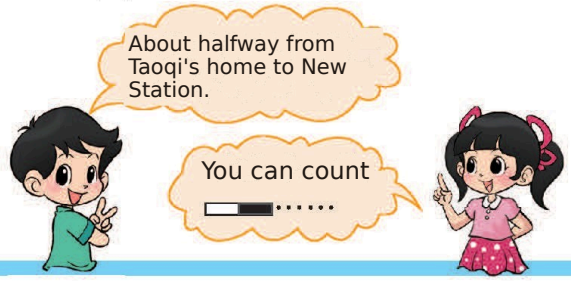
Taoqi goes to Grandma's: 4 hours by train, a change at New Station, then 2 hours by bus.



3 students drew diagrams of the problem. Can you understand them? Talk with a partner.



Taoqi sets off by train at 9:00. About where is the train 2 hours later? Mark it on the map.



How many kilometres is it from Taoqi's home to Grandma's in total?

115 x 4 =

Answer: _____

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Practice

1. Lele's family is going to Old Town. From Guangming Town they take a bus for 3 hours, change at New Town, then a train for 5 hours.

(1) What maths information do you know? Talk about it and draw it.

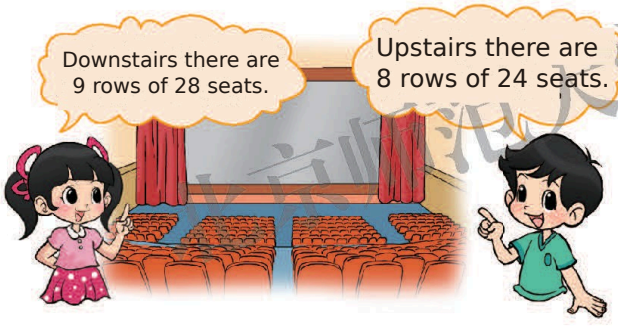
(2) How far is Guangming from New Town?

(3) How far is Lele's home from Old Town?

2. Calculate using the column method.

48×3 32×5 256×3 243×4

3. Going to the cinema.



(1) Grade 3 students fill 5 rows downstairs, with 9 left over. How many students are there in Grade 3?

(2) Grade 4 has 185 students in total. Can they all sit upstairs?

4. Uncle Zhang lives in Beijing. At Spring Festival he goes home to Qingdao by train, via Tianjin and Jinan.

Beijing–Tianjin	122 km
Beijing–Jinan	426 km
Jinan–Qingdao	393 km

(1) How far is it from Tianjin to Jinan?

Draw and calculate.

(2) How far is it from Beijing to Qingdao?

(3) The train leaves Beijing, travelling 152 km per hour on average. About where is it after 3 hours? Mark it on the map.

0×5=?

0×5=? Explain your thinking.

0 added 5 times makes 0, so 0×5=0.

$3 \times 5 = 15$
$2 \times 5 = 10$
$1 \times 5 = 5$
$0 \times 5 = 0$

0 times any number is 0.

Here are some calculations with 0. Work them out using the column method.

$240 \times 2 =$　　$130 \times 5 =$　　$203 \times 3 =$　　$208 \times 7 =$

Write it like this

$$\begin{array}{r} 240 \\ \times\ \ 2 \\ \hline 480 \end{array}$$

$$\begin{array}{r} 2\ 4\ 0 \\ \times\ \ \ 2 \\ \hline \square\square\square \end{array}$$

$$\begin{array}{r} 130 \\ \times\ \ 5 \\ \hline 650 \end{array}$$

$$\begin{array}{r} 1\ 3\ 0 \\ \times\ \ \ 5 \\ \hline \square\square\square \end{array}$$

$$\begin{array}{r} 2\ 0\ 3 \\ \times\ \ \ 3 \\ \hline \square\square 9 \end{array}$$

$$\begin{array}{r} 2\ 0\ 8 \\ \times\ \ \ 7 \\ \hline \square\square\square 6 \end{array}$$

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Calculate. What do you notice?

$0 \times 50 =$　　$0 \times 0 =$

$0 + 126 =$　　$99999 \times 0 =$

$0 + 0 =$　　$78 + 0 =$

0 times any number is 0, and 0 plus any number...

Practice

1. Use examples to explain why 0 times any number is 0.

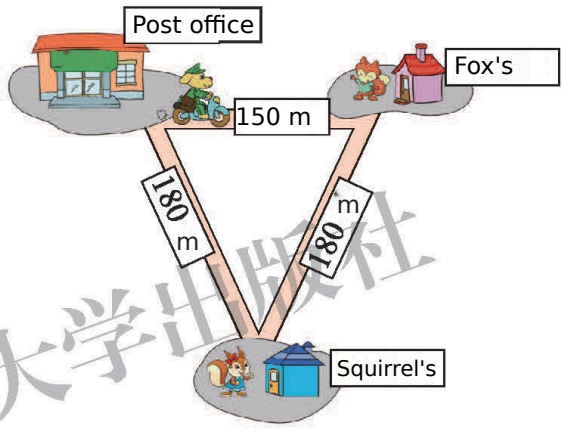
2. Calculate using the column method.

640×2	302×5	105×3	205×4
604×2	320×5	150×3	250×4

3. The puppy delivers letters.

(1) The puppy takes a letter from the post office to the fox. At least how many metres must it walk there and back?

(2) The puppy takes a letter to the fox, then one to the squirrel, then goes back to the post office. At least how many metres must it walk?



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4. Fill ◯ with ">", "<" or "=".

105×6 ◯ 600	190×5 ◯ 1000
180×4 ◯ 800	140×7 ◯ 980

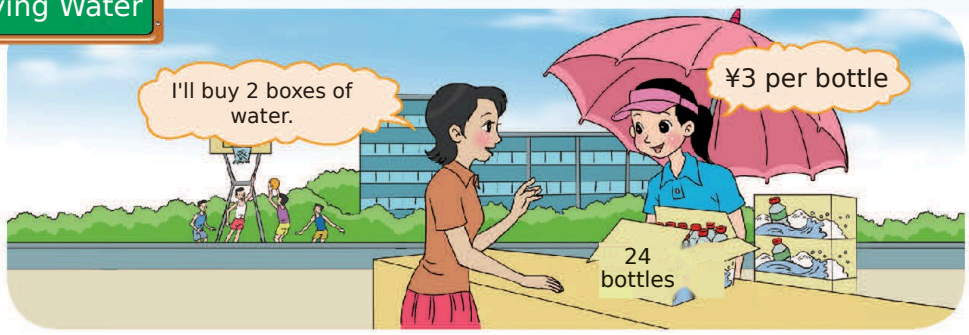
5. The table shows the specimens the Grade 3 classes collected on Science Day.

Class / Type	Class 3 (1)	Class 3 (2)	Class 3 (3)	Class 3 (4)	Total
Insects	105	105	105	95	
Plants		67	67	67	280

(1) How many plant specimens did Class 3 (1) collect?

(2) How many insect specimens did Grade 3 collect in total?

Buying Water



Estimate: is ¥150 enough?

24×2 is less than 50, and 50×3=150, so ¥150 is enough

Work out how much Teacher Zhang spent in all.

First find how much one box costs, then...

$24\times3=72$ (yuan)

$72\times2=144$ (yuan)

It can also be written as one number sentence.

$24\times3\times2$

$=\square\times2$

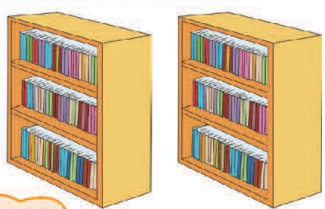
$=\square$ (yuan)

Is there another way?

Answer: _____

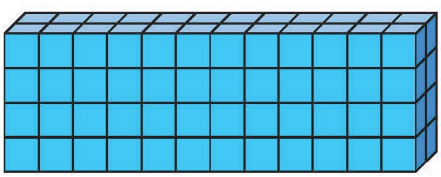
Say what the pictures show, then calculate.

How many books in all?



Each shelf holds 35 books.

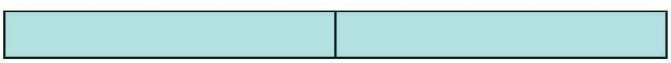
How many blocks in all?



Chickens 45

How many geese?

Ducks

Geese 

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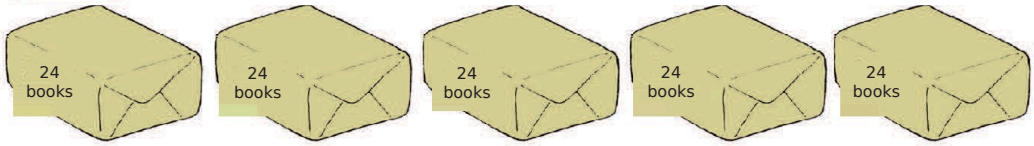
Practice

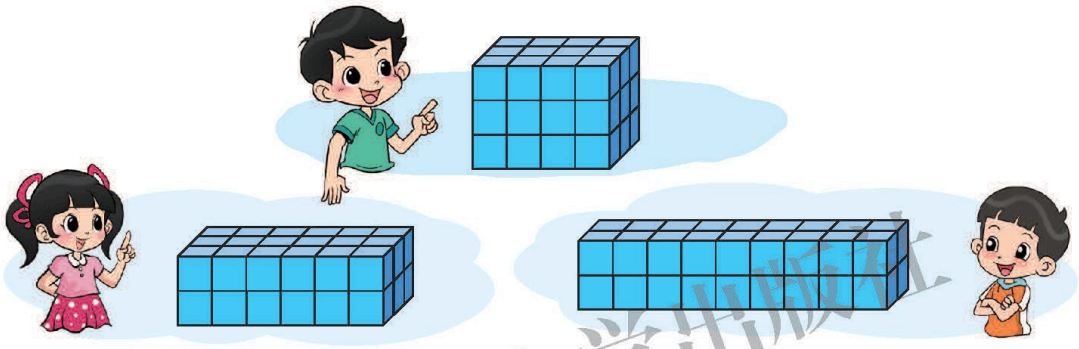
1.



(1) Each book costs ¥4. Estimate: is ¥500 enough to buy 5 packs of books?

(2) Calculate: how much money is needed in all?

2. How many blocks did they each use?

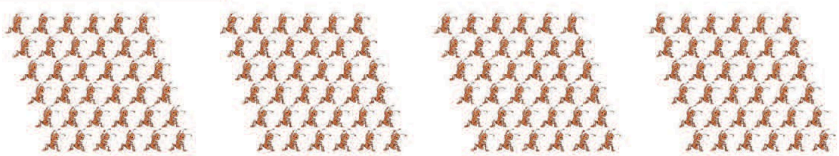


3.

$15 \times 6 \times 8$	$3 \times 9 \times 2$	$(15 + 35) \times 6$
$32 + 17 \times 3$	$5 \times 18 \times 7$	$105 \times (2 \times 5)$

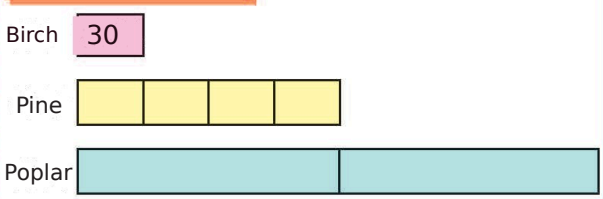
4. Talk about the information in the pictures below, then calculate.

How many ants in all?



How many pigeons in all?

How many poplars?



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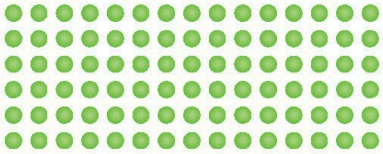
Exercise 5

1. Circle and calculate.

$15 \times 6 = \square$



$24 \times 4 = \square$

2. Fill in.

$54 \times 4 = \square$

×	50	4
4		

$\square + \square = \square$

$91 \times 9 = \square$

×		

$\square + \square = \square$

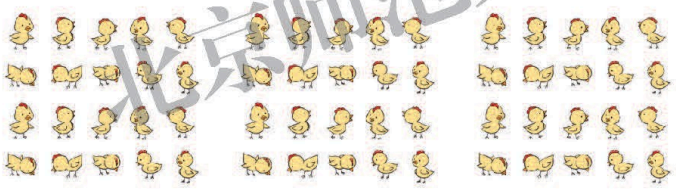
$102 \times 7 = \square$

×	100	2
7		

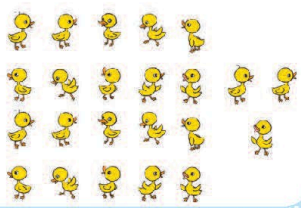
$\square + \square = \square$

3. Calculate and explain.

How many chicks in all?



How many ducklings in all?



4.

300×6	400×7	500×3
15×6	7×7	40×3
315×6	407×7	540×3

5. Use the column method and explain each step.

$216 \times 4 =$

$53 \times 8 =$

$490 \times 3 =$

$607 \times 5 =$

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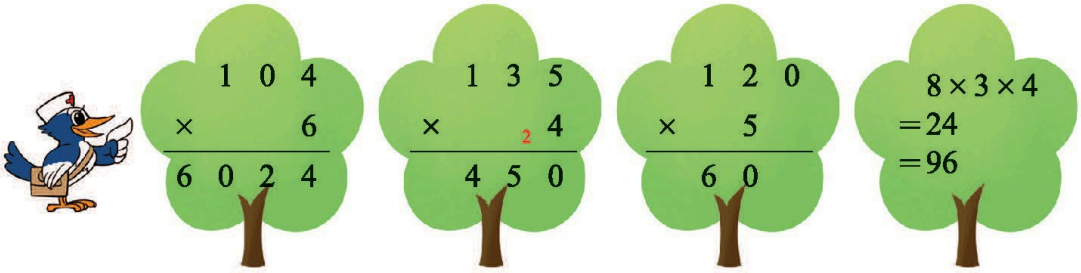
6. Forest doctor.

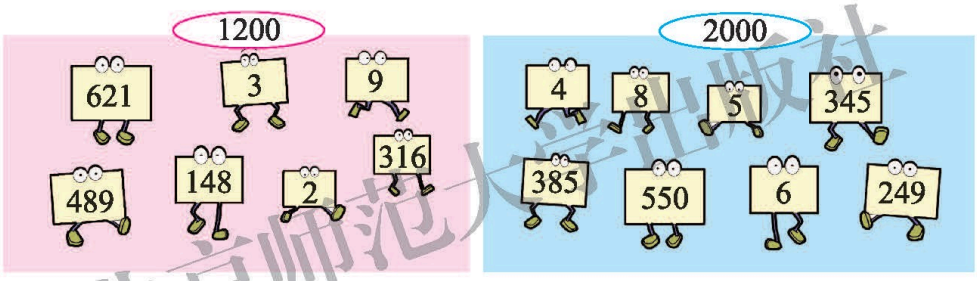


7. Calculate and compare. What do you find? Can you write two more pairs like these?

$12 \times 6 \bigcirc 16 \times 2$ $25 \times 4 \bigcirc 24 \times 5$

$14 \times 5 \bigcirc 15 \times 4$ $16 \times 5 \bigcirc 15 \times 6$

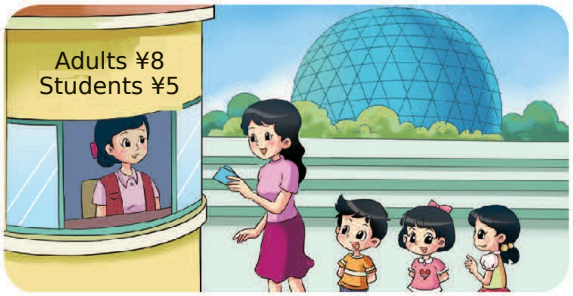
8. Which two numbers have a product closest to the number in the circle? Match them.



9. Can all these pumpkins be taken away in one trip?

10. 5 teachers from Shengli Primary School take 88 students to the science museum. Is ¥500 enough for the tickets?

How much did they actually spend?



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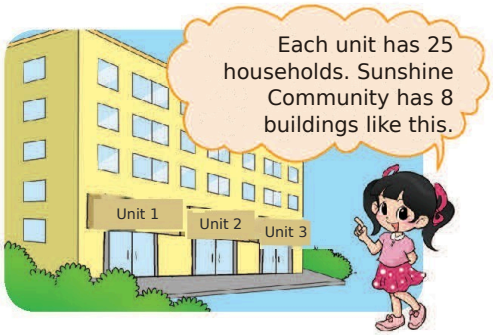
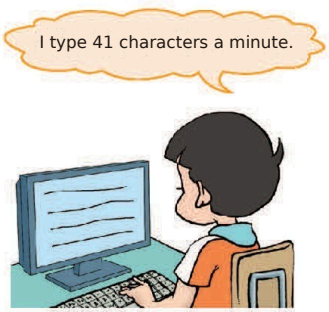
11. (1) How much do Qisi's family need to pay for the tickets?

(2) How many households are there in Sunshine Community?



12.

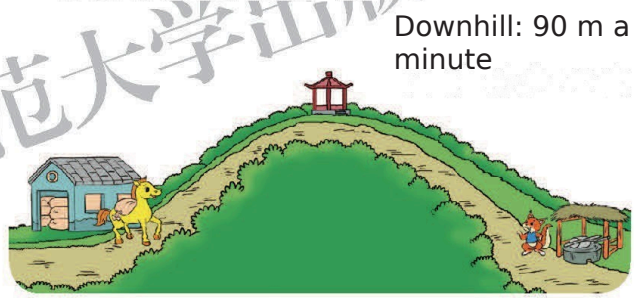


(1) How many characters can Qisi type in 5 minutes?

(2) An article has 325 characters. Can Qisi type it in 7 minutes?

(3) Miaoxiang types 6 more characters per minute than Qisi. Can she type this article in 7 minutes?

13. Every day, the little horse leaves home carrying grain to the mill. First it goes uphill for 9 minutes, covering 630 m, then goes downhill for 8 minutes to reach the mill.



(1) What maths information do you have? Talk about it and draw it.

(2) The horse leaves at 8:00. About where is it after 5 minutes? Mark it on the picture.

(3) How many metres does the little horse travel from home to the mill each day?

14. Calculate. What do you notice?

Write two similar number sentences and calculate, then discuss the reasoning with your partner.

2×99	2×999
3×99	3×999
4×99	4×999
5×99	5×999

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Unit 7 Years, Months and Days

Calendar

Say what you know about years, months and days.

A year has 12 months...

My birthday is 20/5. This year's Spring Festival is...

Look at Appendix 1 and record the number of days in each month of 2013–2016 in the table. What do you find?

Month	1	2	3	4	5	6	7	8	9	10	11	12
2013												
2014												
2015												
2016												

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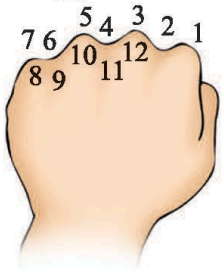
Months 1, 3, 5, 7, 8, 10 and 12 all have 31 days. Months 4, 6, 9 and 11 all have 30 days. February does not always have the same number of days.

A month with 31 days is a long month; a month with 30 days is a short month. A year in which month 2 has 28 days is a common year; a year in which month 2 has 29 days is a leap year.

Look, remember and talk about it with your partner.

Knuckle method



Rhyme method

January, March, May, July, August, October and December always have 31 days; April, June, September and November have 30 days; February has 28 days in a common year, and in a leap year February gets one more day.

In the rhyme, 'La' usually means the twelfth lunar month; here it stands for December. 'Dong' usually means the eleventh lunar month; here it stands for November.

Try It

Use Appendix 2 to record the number of days in February (month 2) for 2009–2016 in the table. What do you find?

Year	2009	2010	2011	2012	2013	2014	2015	2016
Days in Feb.								

Usually there is a leap year every 4 years.

2012 is a leap year, and so is 2016...

Colour the leap years in the table below.

2016	2017	2018	2019	2020	2021	2022	2023	2024	2025

Is the year you turn 18 a common year or a leap year?

How many days are in a common year? And in a leap year?

Practice

1. Find these in this year's calendar.

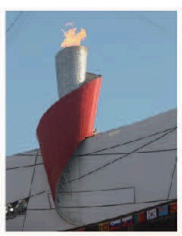
(1) Children's Day is on ____ (month) ____ (day), a ____ (weekday).
(2) National Day is on ____ (month) ____ (day), a ____ (weekday).

The People's Republic of China was founded in 1949. On month 10, day 1 this year, it celebrates its ____th anniversary.

(3) Circle your mum's and dad's birthdays on the calendar in colour. Say whether each is in a long month or a short month.

2. Fill in.

The 29th Summer Olympic Games opened in Beijing on ____ (year) ____ (month) ____ (day). They lasted 16 days and closed on ____ (month) ____ (day).



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3. How many leap years are there from 1996 to 2010? Find and write them.

4. Talk about it.

(1) Is this year a common or a leap year?

(2) The first half of this year has ___ days, and the whole year has ___ days.

5. Find out which month the most classmates were born in, and make a calendar for that month of this year.

( ) year ( ) month						
Sun	Mon	Tue	Wed	Thu	Fri	Sat

(1) Is this a long or a short month? How many Sundays?

(2) Next month starts on ____ (month) ____ (day), a ____ (weekday).

(3) Mark on it the birthdays of classmates born this month. Compare: who is the oldest, who is the youngest, and how many days apart?

6. Guess the birthday.

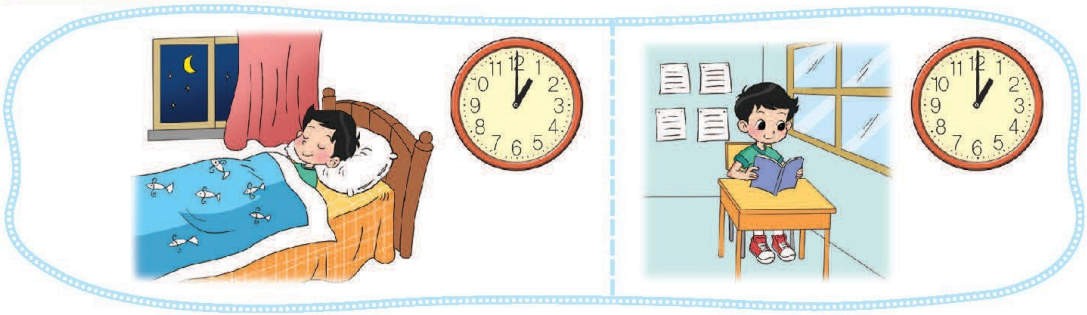
(1) Play the birthday guessing game with your partner.

(2) Qisi had only 3 birthdays by the time he turned 12. His birthday is ____ (month) ____ (day).

Did You Know?

The 365 days we usually call a year is really an approximate number; the exact length is 365 days 5 hours 48 minutes 46 seconds. For convenience, people set a year at 365 days and call it a common year. This way, every four years the calendar falls nearly 1 day behind the exact time. To make up the difference, 1 extra day is added to month 2 of that year, making it 366 days; such a year is called a leap year. It is also ruled that a year ending in two 0s is a leap year only if it can be divided exactly by 400. For example, 2000 is a leap year, but 1700 is not.

Time in a Day



Look and say.

At 1 a.m., Taoqi is asleep.

At 1 p.m., Taoqi is...

1 p.m. can be written as 13:00.

Read and say.

11 p.m. is 23:00.

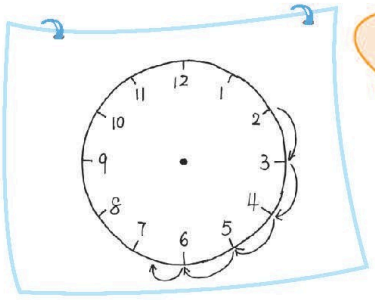
15:00 is 3 p.m.

Fill in and say.

Collection times: 1st 8:40, 2nd 14:00, 3rd 18:30

The interval from the second collection to the third collection is __ h __ min.

14:00 is 2 p.m., so 18:30 is ..... p.m.



15, 16, 17, 18: that's 4 hours. Plus 30 minutes is 4 hours 30 minutes.

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Practice

1. (1) 2 p.m. is ___:00, and 8 p.m. is ___:00.

(2) 23:00 is ___ p.m., and 19:40 is ___ p.m.

2.

The football match starts at 14:10 and lasts 1 h 45 min. It ends at ___:___.

The library is open for ___ h ___ min each day.

3. Draw the hands.

6 : 30

15 : 00

16 : 30

21 : 00

4. Look and fill in.

(1) The train leaves at ___:___ p.m. Ticket checks stop at ___:___.

Checks stop 5 min before departure.

K088441 Tianjin

Tianjin–Beijing Train T554

Departs 18:08 on 06/08/2002, car 02, seat 057. Hard seat, full fare ¥35.00. This train and day only; valid for 2 days.

(2) It takes Xiaoming 20 min to get from home to the ticket gate at the railway station. To catch the train, he must leave home by __:__ p.m. at the latest. Draw and calculate.

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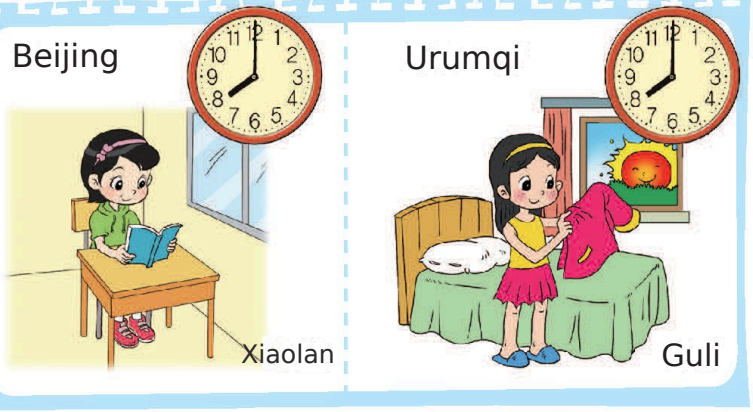
Timetable



Sunrise in Xinjiang is about 2 hours later than in Beijing.

Beijing Qiming School Timetable

Period 1:	8:00 ~ 8:40
Period 2:	8:50 ~ 9:30
Eye exercises:	9:30 ~ 9:35
Break exercises:	9:45 ~ 10:05
Period 3:	10:05 ~ 10:45
Period 4:	10:55 ~ 11:35
Lunch break:	11:35 ~ 13:00

……

Xinjiang Minzu School Timetable

Period 1:	10:00 ~ 10:40
Period 2:	10:50 ~ 11:30
Eye exercises:	11:30 ~ 11:35
Break exercises:	11:45 ~ 12:05
Period 3:	12:05 ~ 12:45
Period 4:	12:55 ~ 13:35
Lunch break:	13:35 ~ 15:00

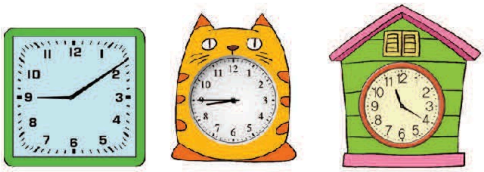
……

Look and say.

At 8 a.m. Xiaolan is in class, while Guli in Xinjiang has just got up...

Guli's first period only starts at 10 a.m., and each period is also 40 min.

In the second period in the morning, Xiaolan and Guli both looked at a clock. Which times on the right might they each have seen?



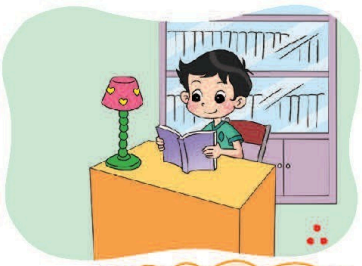
Complete the timetable using what Xiaolan and Guli say.

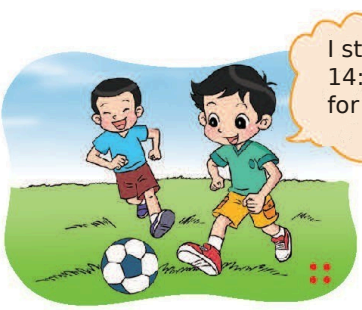
In the afternoon we have two periods from 13:00. After Period 5 there is a 1-hour break, then Period 6.

In the afternoon we have two periods starting at 15:00, then 1 hour of extracurricular activities.

Practice

1. Below is Taoqi's schedule of activities for one Saturday.





(1) Taoqi spent ( ) min on homework.
Draw and calculate.

(2) What is Taoqi doing at these times? Talk about it.

Breakfast: 8:00 ~ 8:30
Homework: 8:50 ~ 9:30
Reading: 10:00 ~ 10:30
Football: ( ) ~ ( )
TV: ( ) ~ ( )

(3) Complete the timetable using what Taoqi says.

2. How do you arrange your time on Saturday? Make a timetable.

____ 's Saturday Timetable

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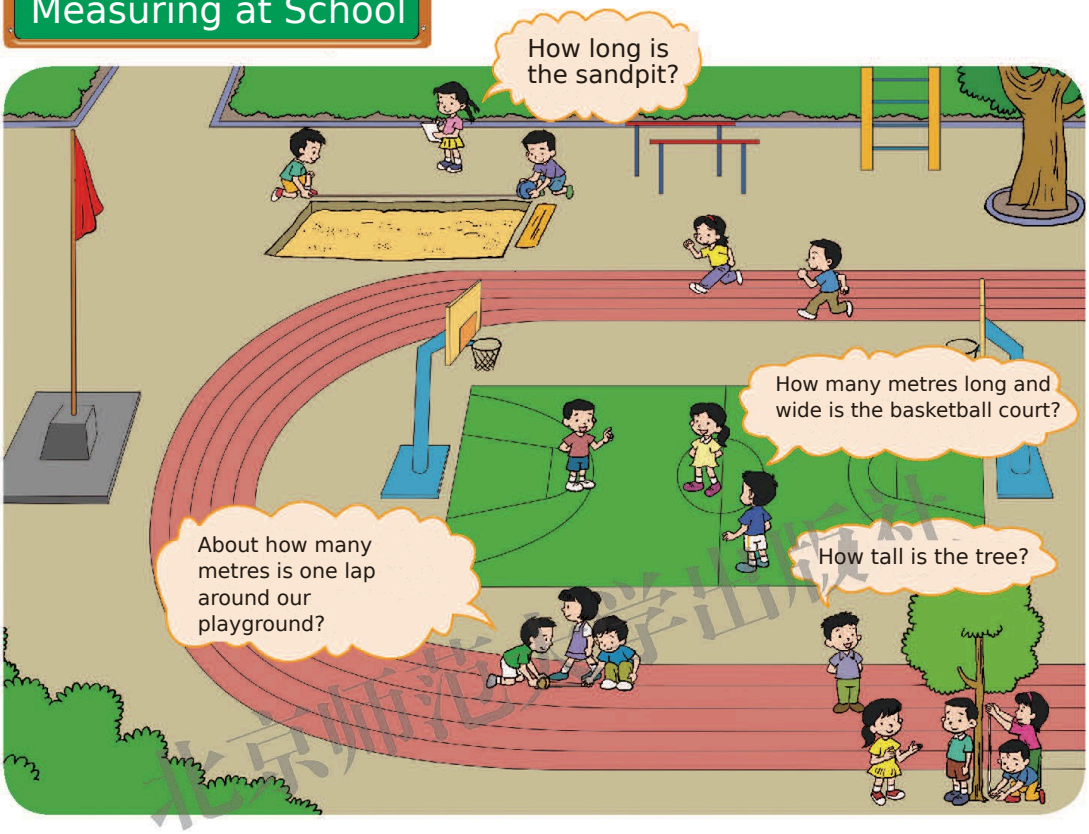
Maths Is Fun

Measuring at School



Task

1. Measure roughly how tall the sports equipment (or a small tree) is, and how long one lap of the playground is.

2. Choose another measuring task you like at school, and do it.

Discuss

1. In your group, discuss possible measuring methods. Which will you use?

2. Which tools will you measure with, and what data will you record?

3. Who will do what in your group?

Do It

Carry out the measurements and fill in the activity record sheet.

_____Group Activity Record Sheet

Date: _____ Members: _____

Object measured	Method	Measuring process	Result
One lap of the playground	Pacing	With a ruler, 1 step is _____ cm…	_____ m

Think

1. In the activity above, which measuring methods did you use? Where else could they be used?

2. Which measuring tools did you choose? What should you pay attention to when using them?

3. What data did you record? Where might this data be useful later?

4. What did you learn from this activity? What else would you like to find out?

Self-Check

How I did in this activity:

I can find a suitable measuring method.			
I can choose suitable measuring tools.			
Our group shares out the work well.			
The results are reasonable.			

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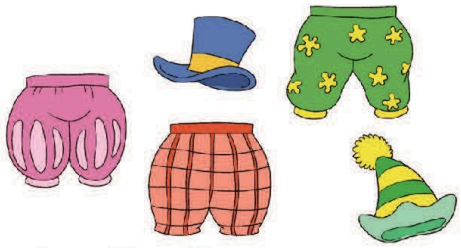
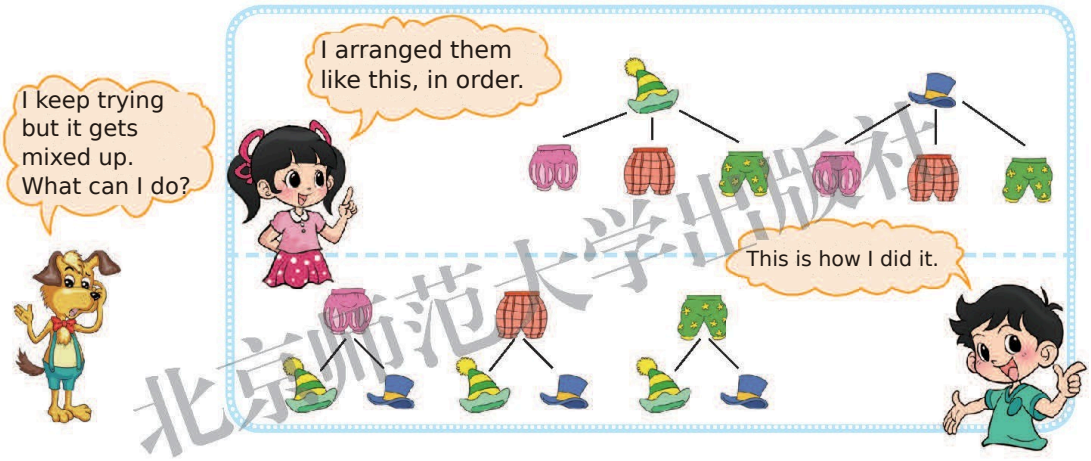
Ways of Matching

Outfits

The circus clown wants to pick a hat and a pair of trousers for his show.



How can they be matched? Arrange them and talk about it.



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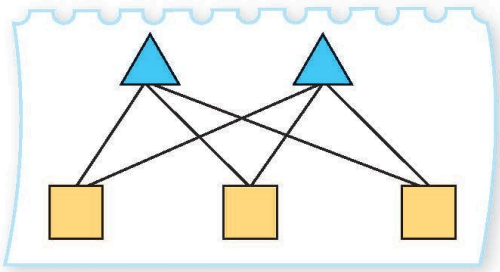
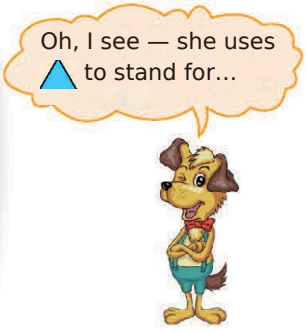
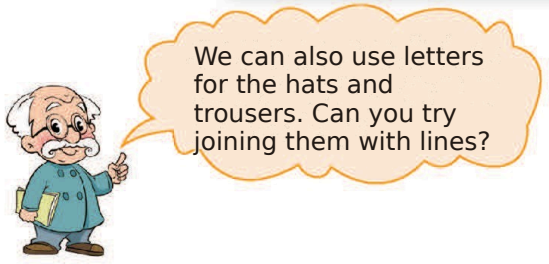
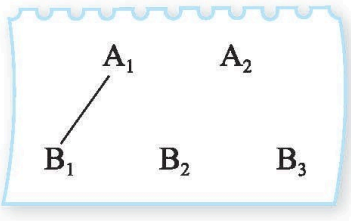
This is how Xiaoxiao shows all the matches. Can you follow it? Discuss with your classmates.

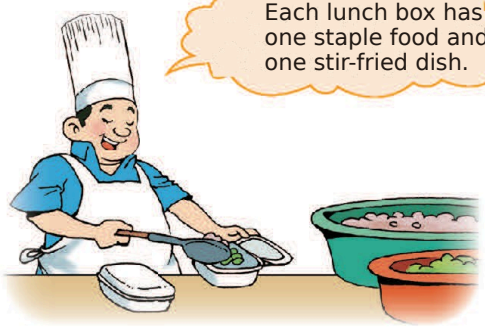








Healthy Meals

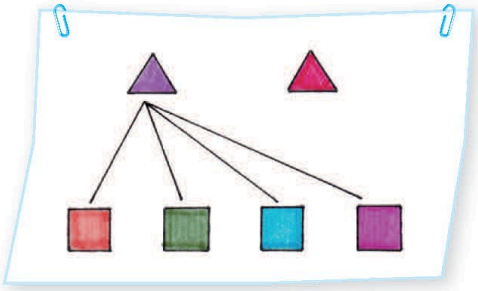


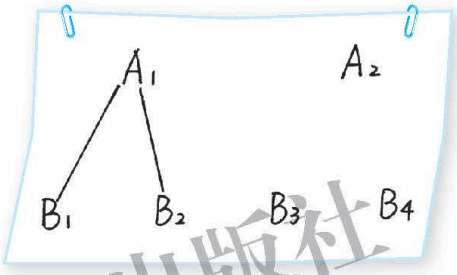
Each lunch box has one staple food and one stir-fried dish.

Today's Lunch

Staple foods: rice, steamed buns.
Dishes: egg and tomato, green pepper and pork
potato slices, braised aubergine

How many different meal combinations are there?





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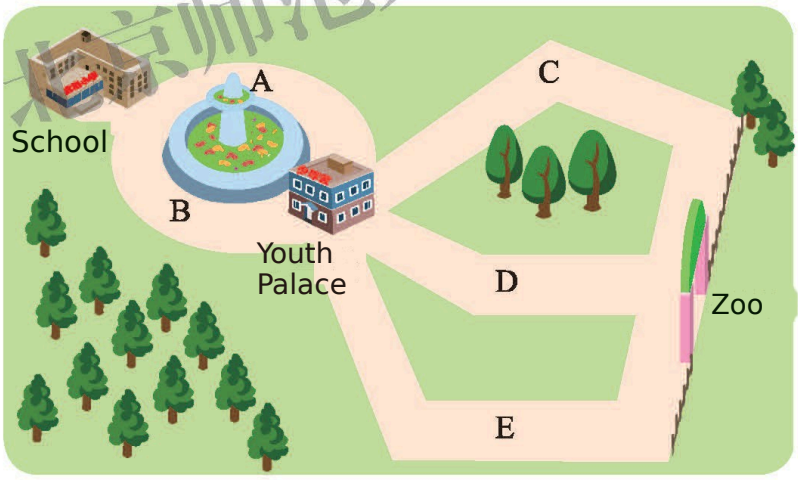
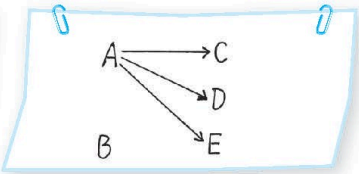
Routes to the Zoo

What other matching problems are there in everyday life?



School → Youth Palace → Zoo: there are ☐ routes in all.

A→C B→C
A→D B
A→E



This diagram looks just like the one for matching outfits!

Time and Maths

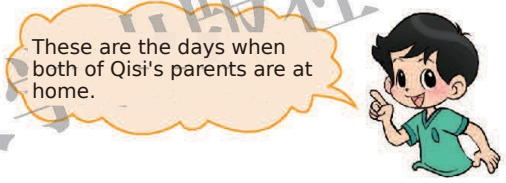
Shared Days Off

Qisi's father is a train driver; he rests for 1 day after every 3 days of work. Qisi's mother is a flight attendant; she rests for 1 day after every 1 day of work. Qisi has Saturdays and Sundays off.

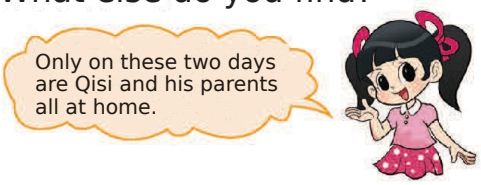
Sept

Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

Mark Father's days off with △ and Mother's days off with ○. What do you notice?



Then mark Qisi's days off with √. What else do you find?



The first picture shows the parents' common rest days. Do you understand it? Show the other rest days in the same way.

4, 8, 12, 16, 20, 24, 28。

Parents' shared days off

Father and son's shared days off

Mother and son's shared days off

Whole family's shared days off

Calendar Patterns

September

Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	

Look at the 4 boxed numbers on the calendar. What do you notice?

12 is 1 more than 11, 18 is 7 more than 11...

I noticed:
$11+19=30$,
$12+18=30$,
...

Try another group.

Look at the 9 shaded numbers on the calendar. What else do you find?

$14-7=7$,
$21-14=7$,
...

$7+15+23=45$,
$14+15+16=45$,
...

Try another group.

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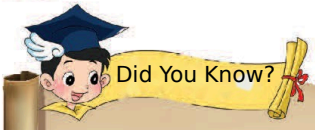
Guess and Say

I chose two dates side by side in a row. Their sum is 25. Do you know which two dates I chose?

I chose two dates one above the other. Their sum is 37. Do you know which two dates they are?



The Story of Young Gauss

Gauss was a great German mathematician of the 19th century, known as the Prince of Mathematics. It is said that as a boy he was given this problem by his teacher: work out 1 + 2 + 3 + 4 + ... + 97 + 98 + 99 + 100. While everyone else was adding step by step, Gauss quickly found the correct answer. Do you want to know his method?

Gauss worked it out like this:

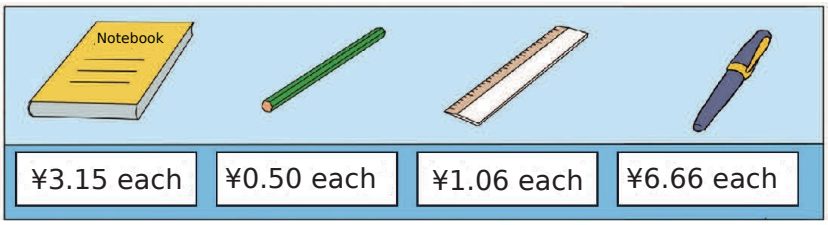
$1+100=2+99=3+98=4+97=5+96=\cdots=50+51=101$

$101\times 50=5050$

Can you understand young Gauss's method? Try it yourself too.

Unit 8 Understanding Decimals

Stationery



Fill in and say.

Notebook _____ yuan = _____ yuan ____ jiao _____ fen
Pencil _____ yuan = _____ yuan _____ jiao _____ fen
Ruler _____ yuan = _____ yuan _____ jiao _____ fen
Pen _____ yuan = _____ yuan _____ jiao _____ fen

Recognise and read.

3.15 Read as: three point one five

0.50 Read as: zero point five zero

decimal point

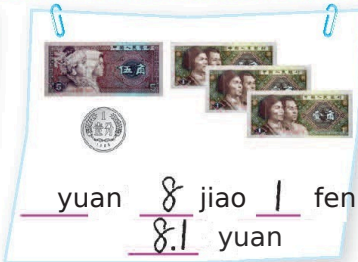
Think and fill in.

_____ yuan ____ jiao ____ fen
_____ yuan

_____ yuan ____ jiao ____ fen
_____ yuan

_____ yuan ____ jiao ____ fen
_____ yuan

_____ yuan ____ jiao ____ fen
_____ yuan



_____ yuan 8 jiao 1 fen
8.1 yuan

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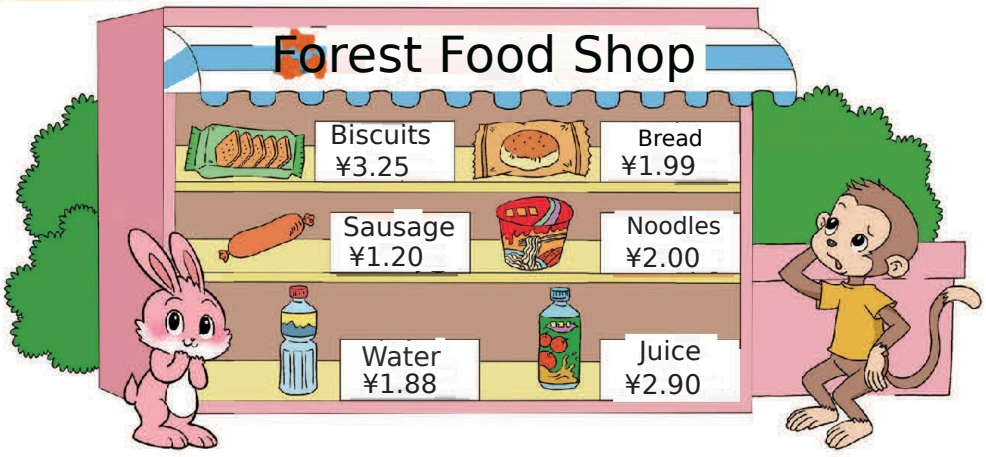
Practice

1.



(1) Little Rabbit buys a packet of biscuits and pays ____ yuan ____ jiao ____ fen.

(2) Little Monkey buys a bottle of juice and pays ____ yuan ____ jiao ____ fen.

(3) Mineral water is ¥1.88. What does each of the two 8s mean?

2. Write and read.

_____yuan

_____ yuan

_____yuan

3. Forest Doctor

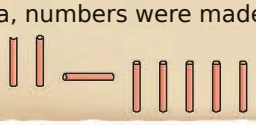
¥7.15 is 7 yuan 15 jiao.

3 yuan 2 fen is ¥3.20.

4 jiao 5 fen is ¥4.5.

The name "decimal" was first used by Zhu Shijie, a Yuan Dynasty mathematician.

In ancient China, numbers were made with sticks; decimal digits went a row lower.

2.15 is shown as:  This is the world's earliest way of writing decimals.

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Shop Around

Look and say.

At which shop is the pencil case cheaper? Explain how you worked it out.

¥4.9 ◯ ¥5.1

¥4.9 =4 yuan 9 jiao =49 jiao,
¥5.1 =5 yuan 1 jiao =51 jiao.

$49 < 51$

……

¥4.9 is less than ¥5, and ¥5.1 is more than ¥5.

……

¥4.9 is just over ¥4 and ¥5.1 is just over ¥5. 4 is less than 5, so ¥4.9 is cheaper.

Taoqi wants to buy an eraser. Which shop is cheapest?

¥1.8 ¥2 ¥1.9

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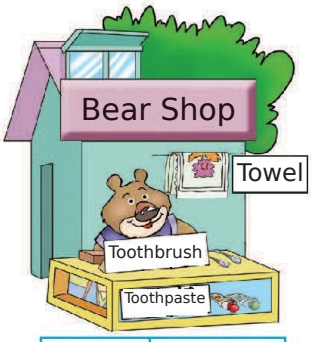
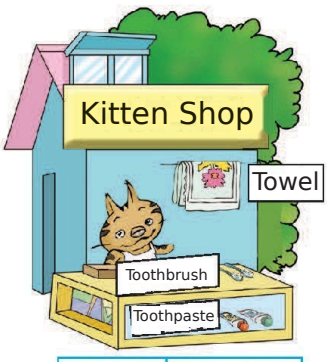
Practice

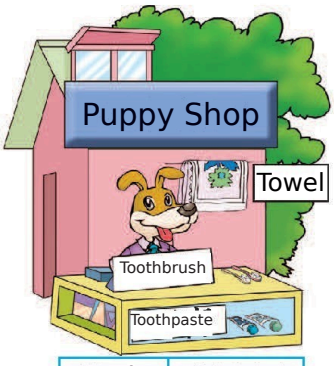
1.



Towel	¥3.80
Toothbrush	¥2.50
Toothpaste	¥4.30



Towel	¥4.00
Toothbrush	¥2.60
Toothpaste	¥4.50



Towel	¥4.10
Toothbrush	¥2.40
Toothpaste	¥3.80

(1) What maths information is in the picture?

(2) Where is the towel cheapest? Explain how you know.

(3) What else do you want to buy? Which shop will you go to?

2. Which is more? Which is less?

___yuan ◯ ____ yuan

___yuan ◯ ____ yuan

3. Fill ◯ with ">", "<" or "=".

¥0.8 ◯ ¥0.6 ¥0.9 ◯ ¥1.1

¥3.3 ◯ ¥3.2 ¥10.9 ◯ ¥11

4. Visit a shop and record the prices of 3 kinds of goods. Compare the price of the same item with your classmates.

Item			
Price			

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Saving Up

Month	Saved
1	¥11.5
2	¥3.2

How much pocket money did Taoqi save in the two months altogether?

□ ○ □ = □ ( )

11 yuan + 3 yuan = 14 yuan

5 jiao + 2 jiao = 7 jiao

so it is ¥14.7.

¥11.5 = 115 jiao

¥3.2 = 32 jiao

```
  1 1 5
+   3 2
-------
  1 4 7
```

147 jiao is ¥14.7.

The decimal points must line up.

```
yuan  jiao
  1 1 . 5
+   3 . 2
-------
  1 4 . 7
```

Answer: _____ .

Taoqi wants to buy a ¥29.9 schoolbag for children in the disaster area. At least how much more does he need to save?

□ ○ □ = □ ( )

```
yuan  jiao
  □ □ . □
- □ □ . □
-------
  □ □ . □
```

Answer: _____ .

Using the questions above, say why the decimal points must line up.

When the decimal points line up, yuan lines up with yuan...

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Practice

1.

¥15.40

¥4.20

(1) How much does Lele pay for these two toys altogether?

Tell us how you worked it out.

(2) How much cheaper is the toy bear than the toy car?

2.

yuan	jiao
1	2
+ 5	7

yuan	jiao
6	8
− 5	1

yuan	jiao
2	2
+ 10	7

yuan	jiao
3	6
+ 6	2

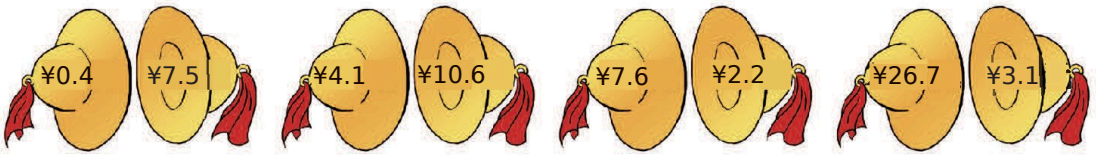
yuan	jiao
3	9
− 2	8

yuan	jiao
17	8
− 0	4

3. Match the Pairs

¥7.5 + ¥0.4

¥7.5 − ¥0.4



4. (1) How much less does one ceramic cup cost than one glass cup?

(2) With ¥20, which two cups could you buy? How much does each choice cost?

(3) If you buy 1 of each kind of cup, is ¥30 enough?

Ceramic cup: ¥11.20 each

Plastic cup: ¥6.10 each

Glass cup: ¥13.60 each

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Posting Books

Estimate: about how much is needed?

¥1.6 is more than ¥1, and ¥12.8 is more than ¥12, so it must be more than ¥13

¥1.6 is less than ¥2, and ¥12.8 is less than ¥13, so it won't be more than ¥15.

Work out how much Miaoxiang must pay in total.

□ ○ □ = □ ( )

1 yuan +12 yuan =13 yuan

6 jiao+8 jiao=14 jiao=1 yuan 4 jiao

13 yuan+1 yuan 4 jiao
=14 yuan 4 jiao

14 yuan 4 jiao is the same as ¥14.4.

¥1.6 =16 jiao
¥12.8 =128 jiao

$$\begin{array}{r} 1\ 6 \\ +\ 1\ 2_{1}\ 8 \\ \hline 1\ 4\ 4 \end{array}$$

144 jiao is the same as ¥14.4.

yuan jiao

$$\begin{array}{r} 1\ .\ 6 \\ +\ 1\ 2\ .\ 8 \\ \hline \end{array}$$

Answer: _____ .

Miaoxiang paid ¥15. How much change should she get?

□ ○ □ = □ ( )

15 yuan -14 yuan =1 yuan

¥1 =10 jiao

10 jiao -4 jiao =6 jiao

6 jiao is...

¥15 =150 jiao

¥14.4 =144 jiao

$$\begin{array}{r} 1\ \dot{5}\ 0 \\ -\ 1\ 4\ 4 \\ \hline 6 \end{array}$$

6 jiao is...

yuan jiao

$$\begin{array}{r} 1\ \dot{5}\ .\ 0\,0 \\ -\ 1\ 4\ .\ 4 \\ \hline \end{array}$$

Borrow ¥1, which is 10 jiao, then 10 jiao minus 4 jiao .....

Answer: _____ .

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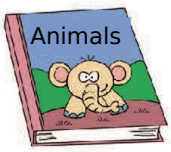
Practice

1.

¥21.80



¥14.50

(1) Qisi wants these two books. Is ¥35 enough?

(2) Work out how much is actually needed.

2.

yuan jiao	yuan jiao	yuan jiao	yuan jiao
1 . 8	1 3 . 2	8 . 3	7 . 4
+ 3 . 5	− 9 . 1	+ 6 . 7	− 5 . 9

3. Forest Doctor

yuan jiao	yuan jiao	yuan jiao
1 0 . 5	2 0 . 3	1 5 . 6
− 3 . 6	+ 4 . 7	− 6
7 . 1	2 4 . 0	1 5 . 0

4.

¥3.80

¥1.20

¥2.50

¥6.50

¥4.80

¥0.80

(1) How much are a packet of biscuits and a bag of seeds in total?

(2) How much more does a bag of jelly cost than a box of crisps?

(3) Qisi has only ¥8. Is it enough for a bag of bread and a bag of jelly?

(4) Miaoxiang has ¥5. Which two foods can she buy? Give two ways.

	Price of item 1	Price of item 2	Total amount	Money left
Way 1				
Way 2				

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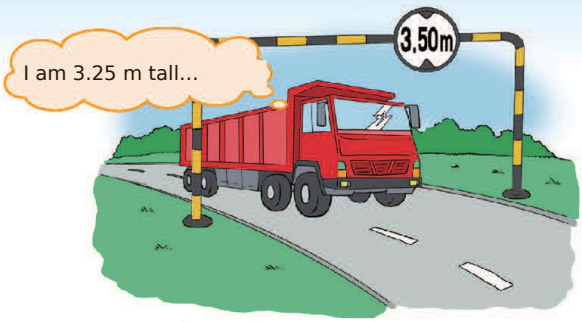
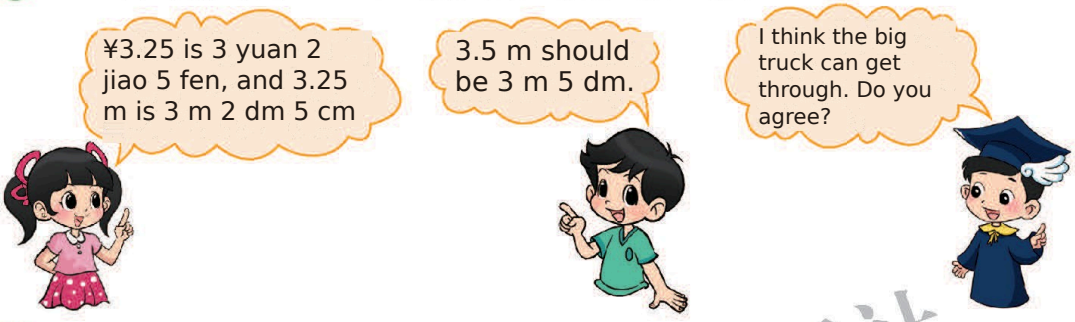
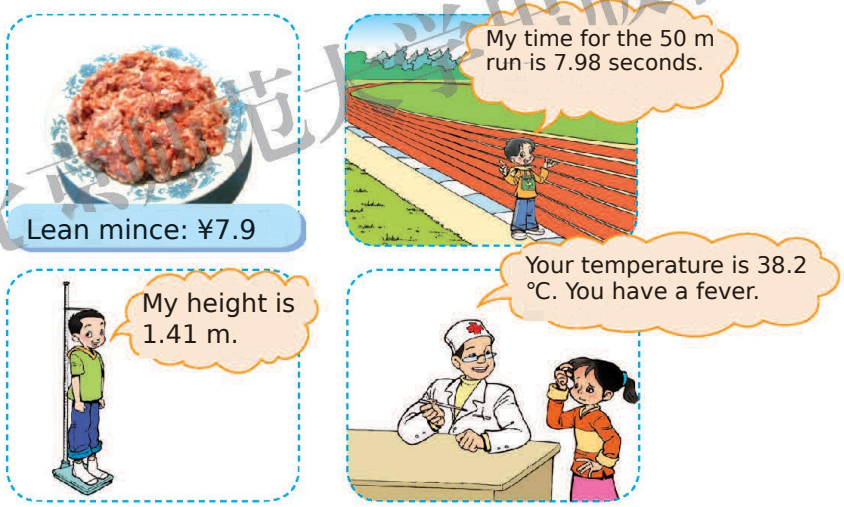
Can It Pass?



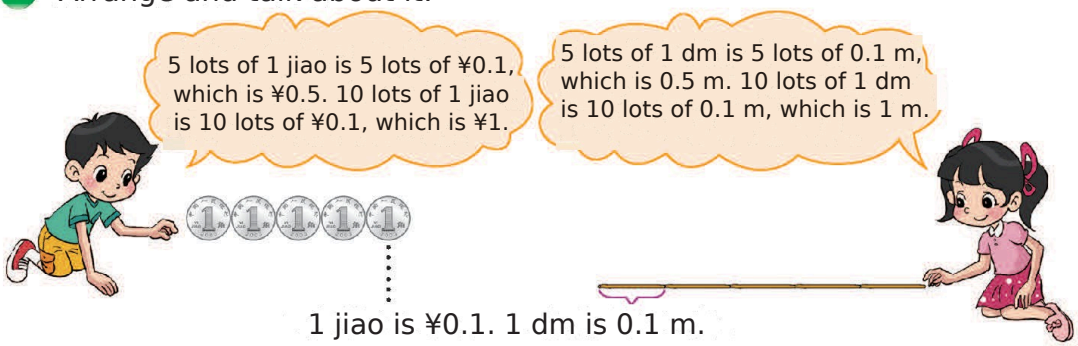
How tall is 3.25 m? What about 3.50 m? Think and talk about it.



There are many decimals in everyday life. Look and read.



Arrange and talk about it.



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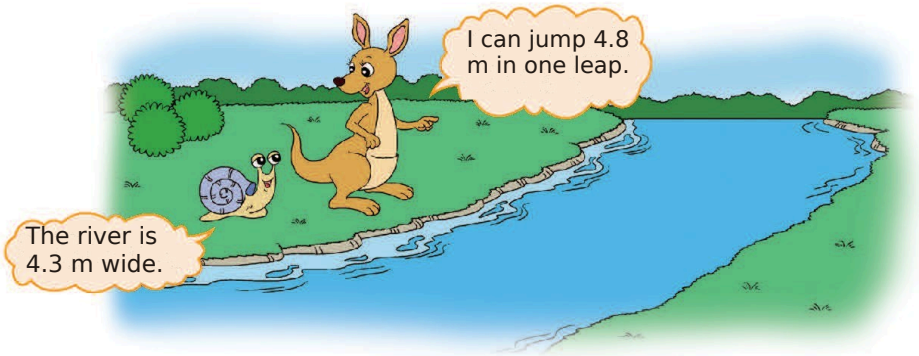
Practice

1. The little kangaroo crosses the river.



(1) How long is 4.3 m? What about 4.8 m? Think and write.

(2) Can the little kangaroo jump across in one leap? Compare and talk about it.

2. Read, then find decimals in daily life with a partner.

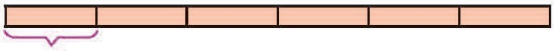
The Chinese white dolphin is a national first-class protected animal, known as the "mermaid" and the "giant panda of the water". Its body can reach 2.7 m in length.

3. Count and write.

(1)

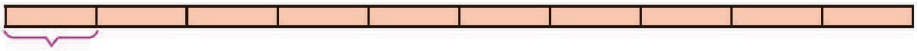
5 lots of 1 jiao is 5 lots of ( ) yuan, which is ( ) yuan.

10 lots of 1 jiao is 10 lots of ( ) yuan, which is ( ) yuan.

(2) 

1 dm

6 lots of 1 dm is 6 lots of ( ) m, which is ( ) m.



1 dm

10 lots of 1 dm is 10 lots of ( ) m, which is ( ) m.

4. How many yuan is 12 lots of ¥0.1? How many metres is 12 lots of 0.1 m?

Exercise 6

1. Write as a decimal

____yuan

____yuan

____yuan

2. Write the price of each item below in yuan, jiao and fen.

¥7.45

____yuan ____ jiao ____ fen

¥1.05

____yuan ____ jiao ____ fen

¥24.8

____yuan ____ jiao ____ fen

¥0.67

____yuan ____ jiao ____ fen

3. In ◯ write "＞" "＜" or "＝"

¥6.8 ◯ ¥6.9　　¥8.1 ◯ ¥7.8　　¥0.40 ◯ 4 jiao

¥5.4 ◯ ¥4.5　　¥20 ◯ ¥19.8　　¥9.9 ◯ ¥10.1

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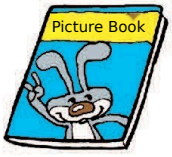
4. What are the prices of the three items below in yuan, jiao and fen? Tell a classmate.

Sticker ¥0.06



Picture book ¥0.60

Fairy tales ¥6.00

5. Calculate.

1.8 yuan +3.3 yuan =

yuan jiao

$$\begin{array}{r} 1.8 \\ +\ 3.3 \\ \hline \end{array}$$

6.5 yuan − 5.4 yuan =

yuan jiao

8.3 yuan +6.5 yuan =

yuan jiao

50 yuan − 18.6 yuan =

yuan jiao

6. A little survey.

Maths book	Chinese book	English book	Art book	Music book
____yuan	____yuan	____yuan	____yuan	____yuan

Say what each book costs in yuan, jiao and fen.

7. Taoqi has ¥15 and buys a pen holder and a pencil sharpener.

(1) How much do these two things cost?

(2) Is the money he has left enough for a diary? Estimate and calculate.

¥6.50

¥3.40

¥4.70

8. Lele recorded his dad's and mum's phone bills for these two months.

	Jan	Feb	Total
Dad	¥30.2		¥61.0
Mum	¥26.7	¥20.4	
Total			

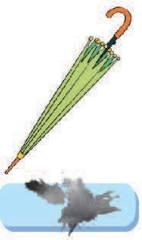
(1) Complete the table.

(2) Find out your family's phone bills for this month and work out the total.

9.

¥0.80

¥2.90



¥3.40

¥20.30

¥15.60

(1) How much do a torch and a battery cost?

(2) I have ¥20. Is it enough for a pair of trainers and a sun hat?

(3) An umbrella costs ¥5.9 less than a watch. How much do an umbrella and a watch cost in total?

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Final Review

Review and Share

Number and Algebra

On your own

1. Use the situations below to explain what each decimal means.

¥2.50

¥3.00

¥2.80

Ribbon: 3.05 m long

Tree: 5.55 m tall

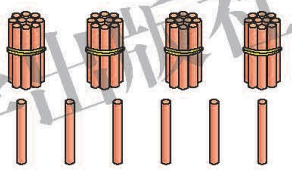
2. Order the prices of the first three items above from lowest to highest. Explain your thinking.

3. Circle, calculate and explain how you did it.

$14 \times 3 =$

$$\begin{array}{r} 1\ 4 \\ \times\quad 3 \\ \hline \end{array}$$

$46 \div 2 =$



4. Calculate, then say the order of operations.

$327 + 46 - 135$ $\qquad$ $(124 - 62) \times 6$

$1000 - 582 - 104$ $\qquad$ $72 \div 8 + 75$

5. Collect the problems you often get wrong and say what to watch out for when calculating.

6. Calculate each of the following and explain why the decimal points must be aligned.

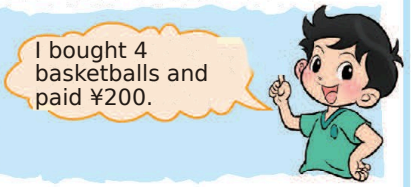
yuan jiao	yuan jiao	yuan jiao	yuan jiao
3 . 4	9 . 8	1 2 . 5	7 0 . 0
+ 1 . 5	+ 4 . 6	− 7 . 4	− 1 7 . 5

7. Solve the problem below, and tell your partner how you solved the real-life problem.

How much change should he get?

¥32

¥45



8. What have you learned about years, months and days? Summarise it and tell your partner.

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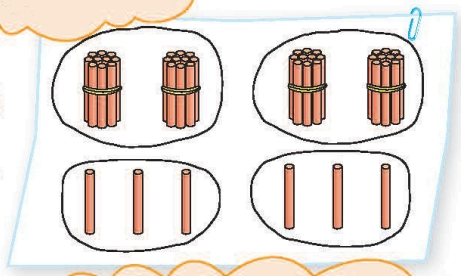
Share ideas

First work out 40÷2=20, then 6÷2=3. The answer is 23.

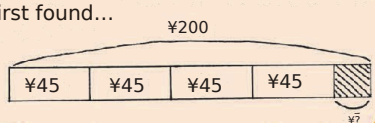


When the decimal points line up, yuan lines up with yuan and jiao lines up with jiao.

yuan jiao

$$\begin{array}{r} 12.5 \\ -\ \ 7.4 \\ \hline 5.1 \end{array}$$

For problem 7, I drew a diagram and first found...



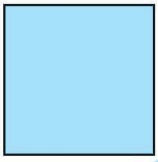
To find the change, subtract the cost of the basketballs from the money paid.

Shapes and Geometry

On your own

1. Choose a desk, the teacher's desk or another object in the classroom. Look at it from different positions and talk with a partner. What do you notice?

2. Explain with an example what perimeter is. How could you measure the perimeter of the shapes below?



3. How do you calculate the perimeter of a rectangle and a square? Measure first, then calculate.

Share ideas

The perimeter of the coin face is the distance once around the coin. Wrap a string exactly once around the coin, then measure the string...

Measure each side of the rectangle and add them up. Or use the rectangle's properties: add the length and width, then multiply by 2.

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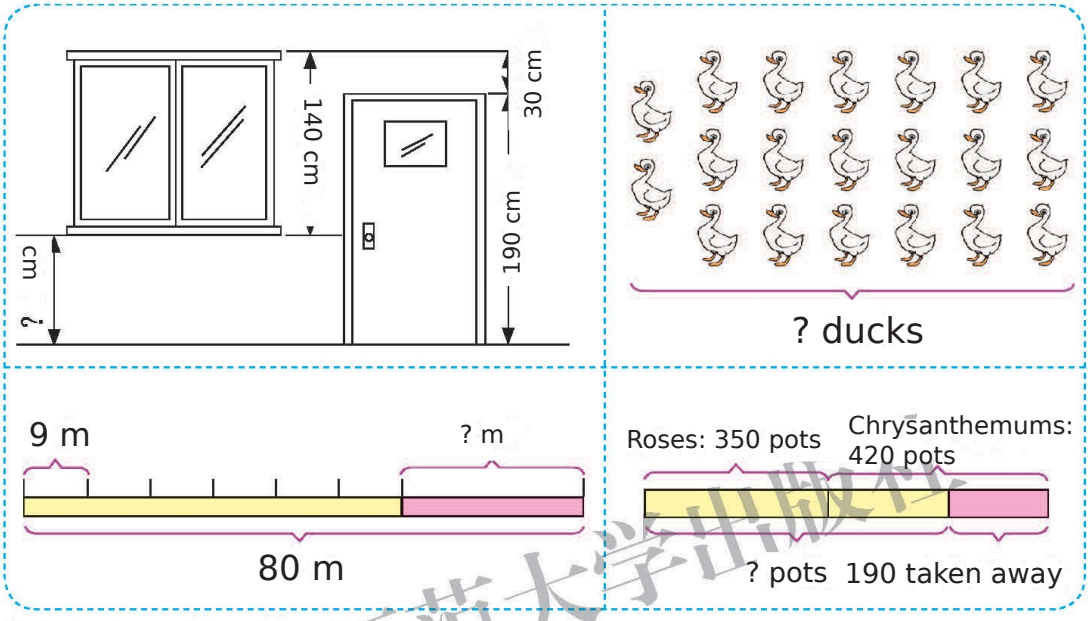
Exercise

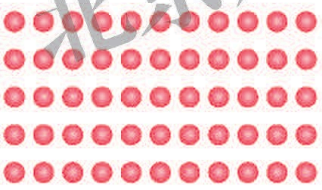
Number and Algebra

1. Discuss and calculate.

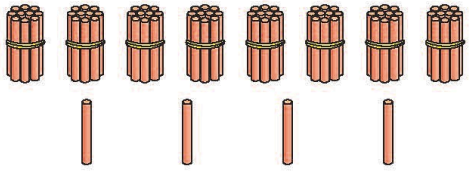


2. Circle and calculate.

$11 \times 5 =$



$84 \div 4 =$



3.

$50 \times 8 =$	$23 \times 3 =$	$320 \div 4 =$	$96 \div 3 =$
$7 \times 60 =$	$18 \times 5 =$	$630 \div 9 =$	$66 \div 6 =$

4. Calculate.

$28 \times 3 =$	$203 \times 4 =$	$347 + 125 - 98 =$
$162 \times 7 =$	$240 \times 5 =$	$950 - 362 - 190 =$
$400 \div 8 - 45 =$	$(92 + 28) \times 4 =$	$12 \times 2 \times 9 =$

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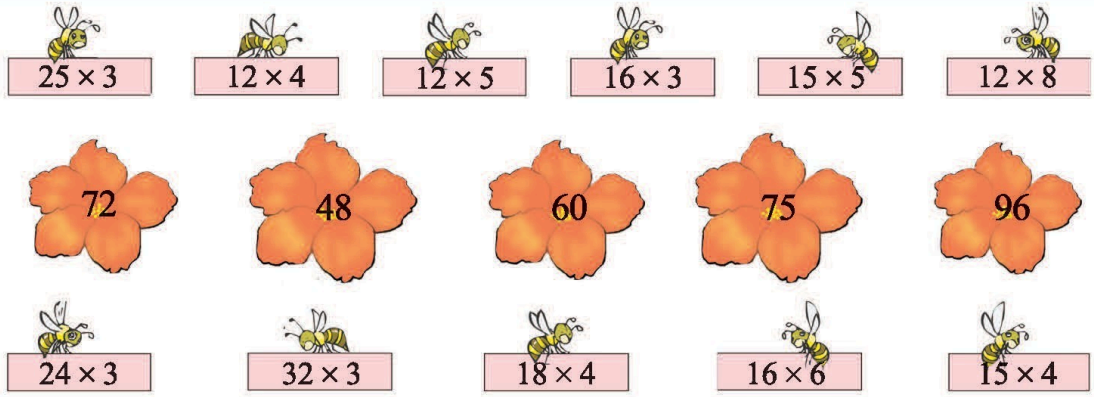
5. Little bees gather honey.



6. Match.

7 p.m.	5 p.m.	12 noon	7 a.m.	12 midnight

7 : 00	19 : 00	12 : 00	17 : 00	24 : 00

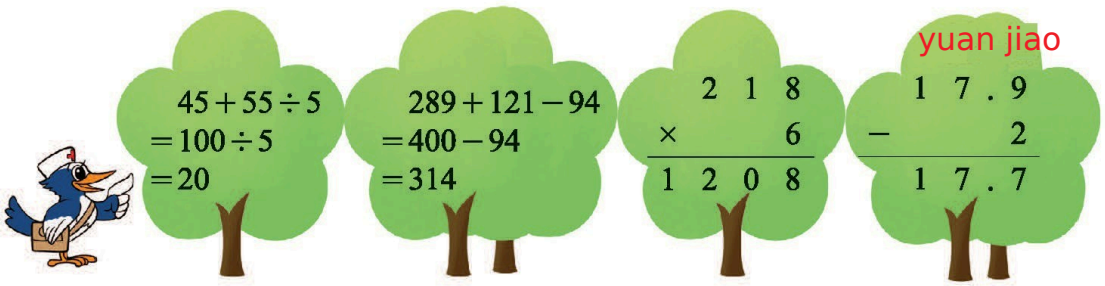
7. Fill ◯ with ">", "<" or "=".

¥1.9 ◯ ¥1.8	¥12.1 ◯ ¥11.9	¥6.5 ◯ ¥6.6
¥4.9 ◯ ¥5	¥10 ◯ ¥9.8	¥35.4 ◯ ¥34.5

8.

yuan jiao	yuan jiao	yuan jiao	yuan jiao
6 . 4	1 . 8	8 . 0	3 . 9
+ 3	− 0 . 6	− 4 . 2	+ 1 . 5

9. Forest Doctor.



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10. To vaccinate the students in three grades, what is the least number of disposable syringes needed?

Grade 1	152
Grade 2	138
Grade 3	147

11. How much more does a bottle of yogurt cost than a bottle of fruit tea?

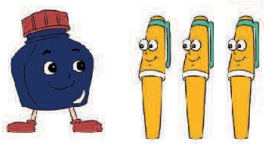
¥6 per bottle ¥24 per box

12.

(1) How much is a bottle of ink?

Spent ¥29 in total

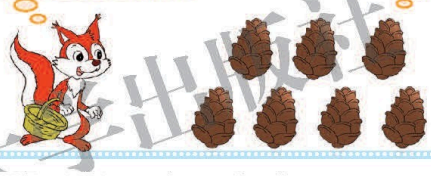
Each pen is ¥6.



(2) How many does each squirrel get?

95 pine cones, shared among 8 squirrels.

7 left over.



13. Below is a table of train distances between the main stations on the "Zhengzhou–Shanghai" line.

	Distance/km
Zhengzhou–Xuzhou	349
Xuzhou–Nanjing	
Nanjing–Wuxi	175
Wuxi–Shanghai	126

(1) How many kilometres is it from Nanjing to Shanghai?

(2) Zhengzhou to Nanjing is 697 km. How far is it from Xuzhou to Nanjing? Draw it on the map above, then calculate.

(3) A train from Zhengzhou travels 95 km per hour. Can it reach Nanjing in 7 hours?

14.

Pay ¥600 first, then pay ¥300 each month for the rest

¥1900

Aunt Zhang buys a digital camera in instalments. In how many months can she pay it off completely?

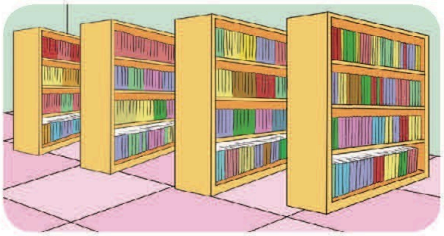
15. The railway from Tianjin to Jinan is 360 km long. On average, how many kilometres does the train travel per hour?

The train from Tianjin to Jinan leaves at 13:30 and arrives at 17:30.

The journey took ( ) hours.

16. How many books are there in total?

40 books per shelf



17. Xiaoxiao's day.

Homework: 9:00 ~ 9:30

Volunteer work: 10:00 ~ 11:00

Lunch: 11:30 ~ 12:00

Games: 14:00 ~ 15:30

Reading: ( )~( )

(1) Xiaoxiao spent ( ) minutes on her homework.

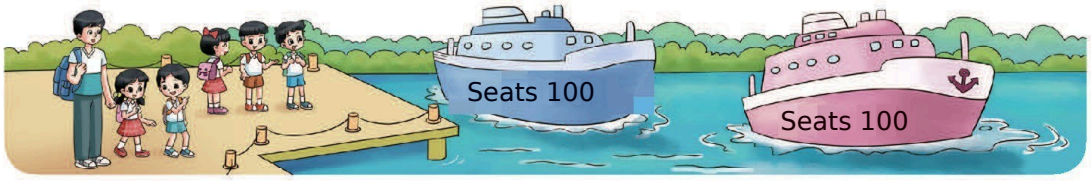
(2) What is Xiaoxiao doing at these times?

Match

10 : 30　　15 : 10　　11 : 45

(3) Xiaoxiao starts reading at 16:10 and reads for 40 minutes. At what time does she finish? Fill in.

18. Grade 3 students rent boats.



Class	1	2	3	4	5	6
Pupils	33	32	33	33	34	34

(1) How many pupils does Grade 3 have?

(2) Which three classes could share a boat? Give a plan.

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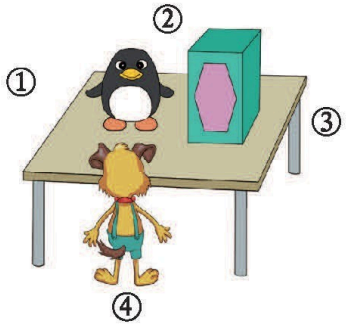
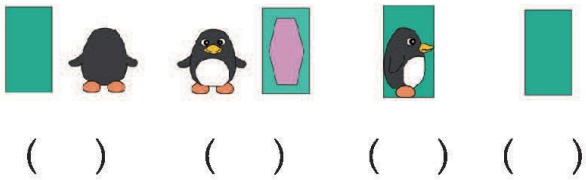
Shapes and Geometry

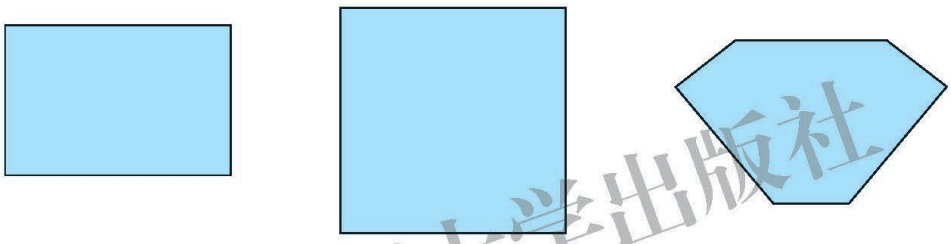
1. Where was Clever Dog standing to see each picture below? Write the number in ( ). See for yourself.



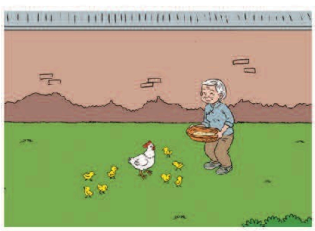


( ) ( ) ( ) ( )

2. Measure first, then calculate their perimeters.



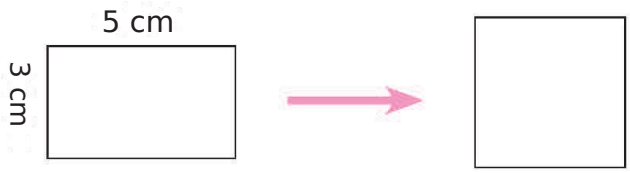
3. Grandma Wang wants to fence off a rectangular chicken pen 5 m long and 3 m wide, using a wall as one side.



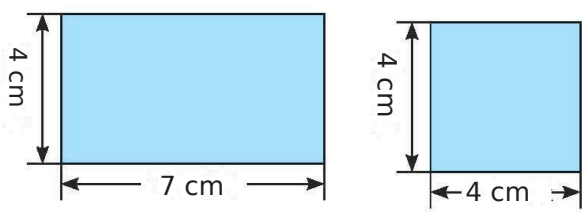
(1) How can she fence it? Draw it.

(2) Find the least fencing needed for each way.

4. Taoqi made a rectangle with wire (as shown below). If the same wire is bent into a square, how many centimetres long is each side?



5. How many centimetres is the perimeter of each shape on the right? If they are joined into one rectangle, what is its perimeter?



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What Did You Learn This Term?

1. _____

2. _____

3. _____

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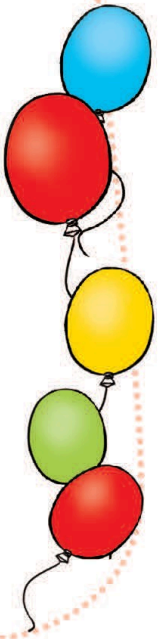

Question Bank

What maths problems have you noticed in your life and studies? Write them down. Can you solve them?

_____

_____

_____

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Appendix 1

January 2013

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Appendix 2

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February 2016

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Afterword

The Compulsory Education Textbooks (Beijing Normal University Edition) were written by many heads and core members of the groups that drew up the national curriculum standards for basic education, together with subject experts, education experts, psychologists and master teachers. They rest on solid research and advanced educational ideas, and offer high-quality writing and professional service. The textbooks strive to reflect the spirit of the national curriculum standards for basic education, make use of many kinds of information resources, show the latest progress in each subject where appropriate, and stress the use of knowledge, skills and ways of thinking in real life. They stay close to students' lives, pay attention to how students learn, meet their different learning needs and promote the all-round development of every student.

The Compulsory Education Textbooks (Beijing Normal University Edition) · Mathematics (Grades 1–6), also known as the New Century Primary School Mathematics Textbooks, fully embody the basic ideas of the new century's reform of the mathematics curriculum, with the healthy growth of children as their fundamental aim. This is the 4th edition of the New Century Primary School Mathematics Textbooks. Building on the strengths of the first three editions, it has been fully revised in accordance with the Mathematics Curriculum Standards for Compulsory Education (2011 Edition).

The new edition has the following distinctive features. Its basic form of presentation is "a situation plus a series of questions", which seeks to make the unfolding of the content, the students' learning, the teachers' teaching and the reaching of the curriculum goals one whole, good for learning and for teaching, so that students keep going through the process of thinking a problem through "from beginning to end". It seeks to give students realistic, interesting and challenging learning materials, to give them time and space to explore and to share their ideas, to show how mathematical knowledge is formed and used, and to meet the needs of different students.

The aim is for students to: appreciate the close links between mathematics, nature and human society; gain the basic knowledge, basic skills, basic ideas and basic experience of activities that they need and that suit their age; develop the ability to find and pose problems and to analyse and solve them; understand the value of mathematics, become more interested in learning it and more confident of learning it well; and gain a first sense of innovation and a scientific attitude.

The chief editors of the New Century Primary School Mathematics Textbooks are Liu Jian, Kong Qiping and Zhang Dan. The other members of the writing group are Hu Guangti, Liu Keqin, Wang Yong, Tao Wenzhong, Wang Mingming, Zhu Dejiang, Chen Xiaomei, Zhang Hong, Qian Shouwang, He Fengbo, Hua Yinglong, Huang Lihua, Zhao Yanhui, Li Tie'an, Wei Huinü, Zhu Yuhong, Ren Jingye and Hou Huiying.

The chief editors of this volume are Chen Xiaomei and He Fengbo. The main people who took part in revising and finalising it were Hou Huiying, Zhu Dejiang, Huang Lihua, Wang Mingming and Xiao Xiaoyu; the final text was compiled and approved by Liu Jian and Zhang Dan. Many local teaching researchers and front-line teachers gave valuable advice on this new round of revision.

Special thanks go to Zhang Zhimeng and other children, and to Zhu Lei, Guo Chenyue and Xu Yue, who were invited to contribute their work.

Because time was short, errors in the textbook are hard to avoid; we sincerely ask our readers to point them out. You are welcome to contact us by phone or letter: Basic Education Branch, Beijing Normal University Press (100875), (010) 58802836, 58802832; Mathematics Studio, Basic Education Curriculum Research Centre, Beijing Normal University (100088), (010) 58435911, xsjmath@126.com.

Beijing Normal University Press